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ENGINEERING THESIS

Electronic and Telecommunication Engineering

**Development of a fiber optic system for spatial localization
of a person in a room**

Katedra Przyrządów Półprzewodnikowych i Optoelektronicznych

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Abstract

This thesis presents the design and implementation of a fiber-optic sensor system for human localization. The motivation for this research comes from the limitations of conventional presence monitoring technologies, such as the lack of stationarity detection in PIR sensors and privacy concerns associated with camera-based systems. The proposed solution utilizes the microbending effect in standard telecommunication optical fibers to detect mechanical pressure caused by human footsteps. The hardware implementation consists of a stable laser transmitter (1550 nm) with Automatic Power Control (APC) and Thermoelectric Cooling (TEC), coupled with a custom 4-channel photodiode receiver. A novel "binary-coded" fiber topology was developed to maximize spatial resolution using a limited number of electronic ports. The system was validated through a fully functional prototype capable of detecting presence and localizing a subject within a defined grid. Experimental results demonstrated that the double-crossing fiber configuration provides the optimal balance between sensitivity and signal stability. The developed device offers a cost-effective, privacy-preserving alternative for smart building applications, particularly in the context of elderly care and fall detection.

Keywords

Presence detection, optical fiber sensor, fall detection

Streszczenie

Niniejsza praca prezentuje proces projektowania i implementacji światłowodowego czujnika lokalizującego osobę w pomieszczeniu. Motywacją do podjęcia badań były ograniczenia konwencjonalnych technologii monitorowania obecności, takie jak brak detekcji obiektów stacjonarnych przez czujniki PIR oraz obawy o prywatność użytkowników związane z systemami wizyjnymi. Zaproponowane rozwiązanie wykorzystuje efekt mikrozgień w standardowych światłowodach telekomunikacyjnych do detekcji nacisku mechanicznego wywołanego przez ludzki krok. Urządzenie składa się z lasera emitującego światło o fali 1550nm, wyposażonego w układy Automatycznej Kontroli Mocy (APC) oraz Chłodzenia Termoelektrycznego (TEC), połączonego z cztero-kanalowym odbiornikiem zbudowanym z fotodiod. W celu maksymalizacji rozdzielczości przestrzennej przy ograniczonej liczbie portów elektronicznych opracowano autorską metodę rozmieszczenia światłowodów opartą na kodowaniu binarnym. System został zweryfikowany poprzez w pełni funkcjonalny prototyp, zdolny do wykrywania obecności i lokalizowania osoby w obrębie zdefiniowanej siatki pomiarowej. Wyniki eksperymentalne wykazały, że konfiguracja podwójnego krzyżowania (double-crossing) zapewnia optymalną równowagę pomiędzy czułością a stabilnością sygnału. Opracowane urządzenie stanowi ekonomiczną i chroniącą prywatność alternatywę dla zastosowań w inteligentnych buynkach, szczególnie w kontekście opieki nad osobami starszymi i wykrywaniu upadków.

Słowa Kluczowe

Wykrywanie obecności, czujnik światłowodowy, wykrywanie upadku

Introduction

The rapid development of the Internet of Things (IoT) and urbanization forces the way modern infrastructure design to be changed. Nowadays, architecture is shifting towards "Smart Buildings," where the primary goal is to achieve minimal energy efficiency and maximum user comfort. Research shows that HVAC (Heating, Ventilation, and Air Conditioning) and lighting systems account for a significant portion of global energy consumption (40%-60% of power consumption in building sector in Europe [1]) and is constantly growing. Consequently, the implementation of intelligent occupancy detection is critical for reducing energy waste by ensuring that resources are only utilized when a human presence is detected within a specific zone. This approach allows to save approximately 15.1% of cooling energy consumption [2].

Beyond energy management, human presence detection plays an important role in healthcare and safety, particularly in the context of an aging society. According to global health reports, falls are a leading cause of accidental injuries and even deaths among older adults [3]. For elderly individuals living alone, a fall can lead to severe consequences if assistance is not immediate. Therefore, there is an increasing demand for Ambient Assisted Living (AAL) technologies capable of continuous, real-time monitoring and fall detection without the need for wearable devices, which users often forget to wear or find uncomfortable [4].

Furthermore, security systems in both residential and critical infrastructure sectors require reliable solutions for intrusion detection. While traditional vision-based detection methods such as cameras provide high accuracy, they often raise significant privacy concerns, especially in private spaces like homes or restrooms [5]. Moreover, standard motion detectors often struggle with blind spots or fail to detect stationary subjects. An ideal detection system should operate autonomously, have a possibility to scale to be used with great area, and respect the privacy while minimizing false alarms in the same time.

In light of these challenges, there is a high demand, and continuously growing, for a non-intrusive, privacy-preserving technology that could be integrated into the building infrastructure to provide reliable and accurate human localization. Therefore, this work proposes using fiber-optic sensor as an alternative to conventional methods, transforming a standard floor into a large-scale, detection surface.

Goal of the thesis

Despite the availability of various human detection technologies, many existing solutions struggle with limitations such as blind spots, high installation complexity, or the need for constant supervision. Addressing these challenges requires exploring alternative sensing mechanisms that can be integrated into the building structure itself.

The purpose of this thesis is to design and develop a fiber-optic based localization system capable of detecting a person within a room while minimizing the need for numerous sensors and complex sensors wiring. The goal is to develop a single unit serving as a controller, light source, and receiver that could cover a large area with sufficient spatial resolution to meet user requirements.

1. State of the Art in Indoor Presence Monitoring

1.1. Key Requirements for Modern Human Detection Systems

To effectively address energy management, healthcare monitoring, and security problems described in the introduction, an ideal detection system must meet specific technical and operational criteria. The most important features include:

- **Energy Efficiency:** With the rising need for sustainable development, the detection systems should be designed to operate under the lowest power consumption possible. Thanks to this approach, the overall carbon emissions decrease. Another advantage is greater lifespan of the batteries of the devices, they can last longer without the need of replacing.
- **High Precision and Reliability:** High precision is essential in human detection devices, both in security and healthcare applications. The system must operate with high-resolution to be able to distinguish between human presence and other environmental factors (e.g., pets, moving curtains). Higher resolution minimizes the rate of false alarms.
- **Automation:** The system should operate without the need of constant human supervision. Automated data acquisition, analysis, and alert generation eliminate human error and speed up the response time in critical situations, such as fall detection.
- **Scalability and Versatility:** To be universal, solution should be easily scalable to be able to cover large areas—such as long corridors or warehouse floors—without the need of installing multiple devices. Furthermore, a single integrated system capable of performing all the tasks (e.g., analysis, detection, communication) is desired because of the easier maintenance and lower cost.
- **Intrusiveness:** Installation should be minimally invasive to the building's infrastructure. System should be as least visible as possible and should not violate people's privacy.

1.2. Overview of Conventional Technologies

Nowadays, monitoring systems use a variety of different sensors and methods to confirm human presence. To evaluate where the strengths and weaknesses of these solutions are, the categorization can be made basing on this features:

- **Cost:** Deployment and maintenance,
- **Power efficiency:** Influences the cost of maintenance,
- **Sensing type:** Active sensing / Passive sensing,
- **Robustness:** False alarm independence and resolution of detection,
- **Privacy concern:** Can it be used in sensitive places,
- **Computational load:** What is needed to process the data.

The most important category listed is **sensing type**. The **active sensing** approach requires external active source – either optical, acoustic or radio-frequency - which transmits some signal into the environment. System requires also the receiver, which reads the reflected or transmitted data and processing unit to analyze the outcomes. The amount of steps and devices influences the power efficiency and cost of the device overall, as the active components are more expensive and more power consuming. Also robustness is endangered, the signal might be corrupted easier (by random interference or intentional jamming). On the other hand, **passive sensing**, bases on signals generated by human body (such as heat or weight), without emitting energy into the room. [6]

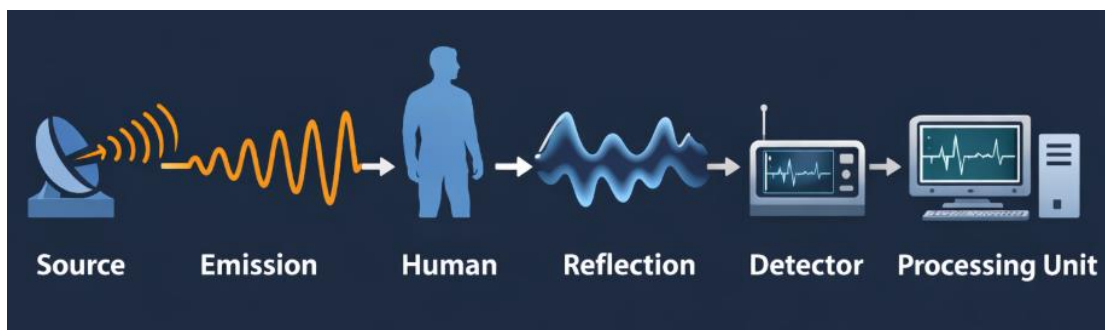


Figure 1 Active sensing operating principle (Source: AI generated)

The most common methods of presence detecting systems are based on electronic sensors and devices, such as [7]:

- passive infrared (PIR) sensors,
- pressure sensors and mats,
- ultrasonic detectors,
- radar-based sensors,
- camera-based vision systems.

These solutions are already well-developed and still under constant improvement, they are capable of achieving high accuracy but need sophisticated algorithms to achieve reasonable power consumption [8]. However, installing such systems are often very costly and requires complex wiring, precise placement of multiple devices to cover the whole area and sometimes constant human supervision. Because of these factors, some solutions becomes hard or even not worth placing with great area.

1.2.1. Passive infrared (PIR) sensors

Operating Principle: PIR sensors operate basing on the pyroelectric effect, where specific crystalline materials generate a transient voltage in response to heating or cooling of pyroelectric material caused by incident infrared (IR) radiation. This temperature variations cause atoms inside the structure to move and polarize. This relation is given by [9]:

$$\Delta P_i = p_i \times \Delta T \quad (1)$$

Where:

- ΔP_i represent the change of component polarization,
- p_i is corresponding pyroelectric coefficient,
- ΔT is a change in temperature.

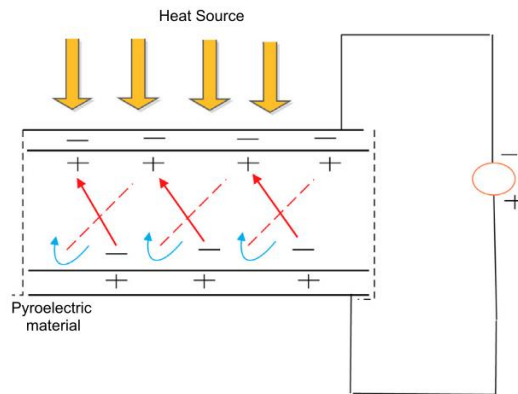


Figure 2 Phenomena of pyroelectricity [9].

To improve detection capabilities, PIR sensors are often coupled with Fresnel lenses that focus incident IR beams into the sensing element. This type of lens is preferred over standard optics because it provides a large aperture and short focal length while maintaining a thin and lightweight profile. Fresnel lens can be shaped to change the detection zone (e. g. narrower, long-range beam, short-range cone). [10]

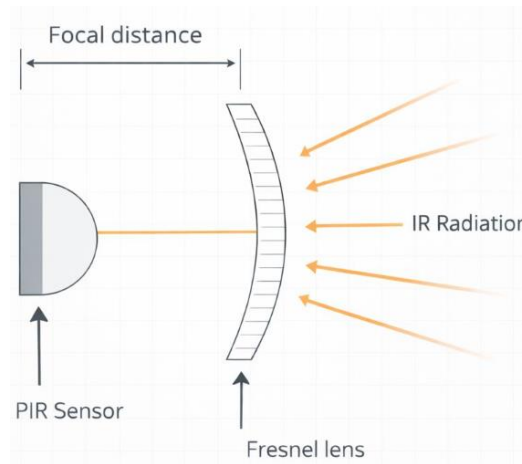


Figure 3 Operating principle of PIR sensor with Fresnel lens. (Source: Own elaboration, generated with AI)

Capabilities and Configurations: While a standard PIR sensor works as a simple binary trigger (movement/no movement), research show that in specific configuration it can extract more sophisticated data such as movement speed, direction, number of people or even person identification. Different study are gathered in Table 1.

Table 1 Results of different measurement approaches using PIR sensor.

Sensing approach	Number of person detected	Achievement	Reference
Single PIR	1	Transverse direction, distance, and speed estimation	[11]
4 PIR + Machine Learning	1	Transverse direction, distance, and speed estimation, person identification	
2 Binary PIR + Booking data + Machine Learning	38	Confirmation of presence while knowing the booking data for specific room	[12]
16 PIR array + 2 ultrasonic + Machine Learning	5	Detection of people with use of CNN algorithm (92% accuracy)	
1 Signal PIR + Machine Learning	14 ± 1	Detection of people with use of iHMM and Poisson Log-Linear algorithms (79% accuracy)	[13]

Limitations: Although research show promising data, it must be remembered that these results were achieved with help of multiple algorithms, additional data and sensors. The amount of additional features required for detection increases computational load and is hardly scalable, to cover greater area it would require hundreds of sensors, wires and powerful processing unit to analyze that data. Therefore, despite their simplicity and privacy preserving sensing mechanism, PIR sensors are not appropriate for use in large area solutions.

1.2.2. Camera based vision systems

Operating Principle: Unlike scalar sensors (e.g. PIR) that operate on 1D data, vision-based systems gather high resolution 2D data and temporal information. Computer Vision (CV) devices capture a video stream, performs background algorithms to isolate moving objects and apply feature extraction techniques (object recognition, human identification, ...). Modern solutions engage Deep Learning models, such as Convolutional Neural Networks (CNN) and You Only Look Once (YOLO), which can simultaneously perform feature extraction and classification tasks [14, 15]. This allows the system to distinguish between humans and false alarms (such as pets).

Capabilities and Advantages: Vision-based monitoring systems gathers the most information out of all detection methods. The most basic overlap with those PIR sensor can achieve: speed, direction, body shape and gait, but can do much more such as:

- **Fall Detection:** By analyzing the body geometry and motion vectors, cameras can detect falls with high accuracy (over 90% [14]),
- **Multi-occupant Tracking:** Advanced algorithms are able to track multiple individuals, even in crowded environment.

Table 2 Analysis of camera-based system.

Camera Type	Sensing Spectrum	Key Advantage	Critical Limitation	Reference
Standard RGB camera	Visible light	High resolution, rich color/texture data for identification	Fails in darkness; Highest privacy violation	[14, 15]
Thermal (IR)	Infrared radiation	Works in total darkness; Better privacy (no facial features)	Lower resolution; Cannot detect cold objects; High cost	[15]
Depth (RGB-D)	Visible + IR Depth	Provides 3D skeletal tracking for fall detection	Limited range; High computational demand	[14]

Limitations and Challenges: Despite their high performance, camera-based systems face critical barriers that stops them from entering the residential and private space market:

- **Privacy Concerns:** The intrusive nature of video surveillance is the most significant drawback. The detailed recording of daily life raises severe privacy issues, often referred to as the "Big Brother" effect. This disqualifies traditional, RGB cameras from being installed in private spaces like bedrooms or restrooms [5]. Additionally, there is a risk that sensitive visual data could be misused by operators or intercepted by unauthorized person.
- **Computational Load and Cost:** Processing high-resolution video stream in real-time requires significant processing unit or cloud processing which leads to costs and privacy concerns [12]. This additional costs often lead to replacing algorithms with old-fashioned human supervision, where operators might miss important data and cause the dangerous consequences. Moreover, the privacy of the occupants is often violated by the operators themselves, who have unrestricted access to the visual data [5].
- **Environmental Sensitivity:** Standard optical RGB cameras require proper lighting conditions to function correctly and fail in darkness. Better here are thermal and Depth cameras which are more immune to that problem but all of them in the same time are prone to occlusion (blind spots) caused by furniture or pillars. To cover a single room fully without blind spots, multiple cameras are often required, which again, significantly increases both the hardware cost and the computational power needed to process the video streams [5, 12].

1.2.3. Floor-Based Monitoring Sensing Technologies

Floor-based monitoring systems allow to transforming any surface into sensor array simply by placing passive devices onto it. Here the division for wearable and non-wearable sensors can be made. Wearable sensors are placed on the clothing, wrists and ankles, on the other hand, non-wearable on are placed mostly on the floor (often referred to as "Smart Floors") and walls [16]. These technologies can be categorized based on their physical properties into:

- Resistive,
- Capacitive
- Piezoelectric.

1.2.3.1. Resistive Sensors

Resistive systems are the most common solution for pressure mats. They utilize Force Sensing Resistors (FSR), which are components whose electrical resistance changes in response to applied mechanical stress.

Operating Principle: Physically, an FSR consist of a conductive polymer film separated from the electrode grid by a thin spacer. Under no-load condition, the sensors act simply as an open circuit ($R > 1M\Omega$), but when pressure is applied, the polymers deform and bridge the electrodes. The resistance drops inversely proportional to the force applied [17]. Mathematically, this is expressed as the conductance:

$$G = \frac{1}{R_{FSR}} \approx k \times (F - F_{TH}) \quad (2)$$

Where:

- F is the applied force,
- F_{TH} is the threshold force
- k is the sensitivity coefficient
- R_{FSR} is the resistance of the FSR

To digitize the signal of an FSR, sensors are typically implemented in a voltage divider configuration. The output voltage V_{out} read by the processing unit is:

$$V_{out} = V_{cc} \times \frac{R_m}{R_m + R_{FSR}} \quad (3)$$

Where R_m is the measuring reference resistor.

This relationship produces a non-linear response, which provides high sensitivity at low forces (e.g. footstep) but saturates at higher weights. The relationship of force an conductance of an FSR is presented in Figure 4.

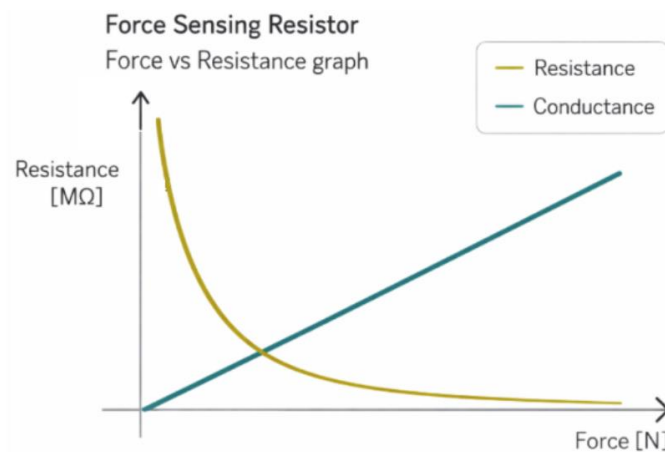


Figure 4 FSR conductance vs resistance graph. (Source: Own elaboration based on equations, generated with AI)

Advanced Configurations: Thanks to their great sensitivity, FSRs are widely used in pressure mats. With sufficient resolution (amount o sensors) such mats are able to detect the center of pressure of standing person, which is often used to improve balance and training applications [18]. They can also track human gait and by analysis, detect gait abnormalities, this is very helpful in making a correct diagnosis and assessing proper rehabilitation for a patient [17, 19]. Closer to this topic, sensing mats are also used in fall detection. Differential sensor configuration cancels out the noise produced by many sensors placed close to each other and is able to detect falls with sensitivity of 89% [20].

1.2.3.2. Capacitive Sensors

Operating Principle: Unlike resistive technologies that rely on direct physical compression, capacitive sensing offers an alternative of detecting human presence through dielectric materials. The fundamental principle of capacitive floor sensing relies on the interaction between the human body and electric filed stored on the capacitor plates [21]. Touching the sensor causes the loss of charge, therefore the change of sensor reading. The capacitance of the sensor can be expressed as:

$$C = \frac{\varepsilon A}{d} \quad (4)$$

Where:

- C Is total capacitance,
- A Is overlapping area of two plates,
- d Is the separation between the plates,
- ε Is the permittivity of the dielectric.

Capacitive sensors typically operate in two models:

- **Loading mode** – When a person stands on a sensor, their body adds a path to the ground, causing the current to be drain from the electrode, the sensor is discharged.
- **Transmit mode** – Additional electrode is placed somewhere in the room and one of them serves as a transmitter. Person standing between them acts as a dielectric and increases or decreases the capacitance.

Capabilities: Capacitive floors have demonstrated high potential for precise indoor localization. For instance, “CapLock” [21] system operating in transmit mode, achieved a localization error for wide-range of trajectories on level of only 0.026m - 0.066m. Moreover, other studies developed a system differentiating walking and falling events analyzing the rate of change in capacitive signals using low-cost planar aluminum electrodes [22, 23].

1.2.3.2. Summary of Floor-Based Sensors

Table 3 presents a comparison of the specific floor-based implementations discussed in this section, highlighting the performance achievements and identification limitations.

Table 3 Analysis of floor-based systems

Technology	Key method	Reported metrics	Critical Limitation	Reference
Resistive [FSR]	Differential sensor pairs	Sensitivity: 89% Specificity: 95% (Fall detection)	1 sensor covers only 3cm ² , 32x32 array used for a whole mat	[20]
Resistive [FSR]	Optimization of track width	Center of pressure error reduced from 17.4% to 5.5%	Significant crosstalk and noise; Expensive removal algorithms.	[18]
Resistive [FSR]	Sensors arrays and analysis	Gait analysis, optimization of sensor placement, footstep force distribution tracking	Large amount of sensors	[17, 19]
Capacitive	Transmit mode for localization	Mean localization error: 0.026m	Sensitive to EMI and stray capacitance.	[21]
Capacitive	Low-cost planar electrodes	Successful fall identification	Range limited to electrode size	[22, 23]

Pressure sensors technique limitations: Although floor-based detection methods seems to work pretty well and could be low-cost while achieving high accuracy, the main disadvantage is their scalability. In presented methods all of them exhibit small spatial resolution – in case of FSR mats it was hundreds of sensors required to cover few meters square [18, 19]. For use in a larger scale, thousands of units would be needed, as well as kilometers of wires to connect them and processing unit able to analyze such amount of data. Additionally, capacitive sensors are prone to EMI and stray capacitance [21], which eliminates this technique from being used in for example factory environment where EMI stays rather high. FSR approach also is prone to interference – placing more sensors close to each other causes crosstalk which might result in “ghost keys”, which means false alarms.

1.3. Limitations of conventional solutions and motivation

As presented in the previous chapters, numerous technologies exist for indoor presence monitoring – ranging from simple PIR sensors to advanced computer vision algorithms. However, no single solution currently satisfies the key requirements defined in Section 1.1. Each technology exhibits significant trade-offs regarding privacy policy, cost, detection accuracy or user convenience.

To provide the general comparison, Table 4 summarizes the characteristics of discussed methods. It also includes Wireless and Radar-based technologies (such as Wi-Fi and RF sensing), which were not analyzed in details in this paper because of their complexity and numerous types of applications and subsections – however, their key parameters are taken into account to ensure complete comparison.

Table 4 highlights a critical gap in the “Non-Wearable” category: systems are either cheap but inaccurate (PIR), accurate but intrusive (Cameras), or privacy-preserving but expensive and hard to scale (Pressure sensors).

Table 4 Summary of electric-based human detection solutions.

Technology	Specific Method	Key Strength	Critical Limitation	Privacy Level
Wearable	IMU / Smartwatch	Portable Low-cost	Compliance issues: users forget to wear or charge the device	High
Ambient (Vision)	RGB cameras	Richest in data (ID, posture)	Privacy violation: “Big Brother” effect; Blind spots	Low
Ambient (Vision)	Thermal cameras	Works in darkness	Economy issues: High cost; Small resolution	Medium
Ambient (Scalar)	PIR sensors	Very low cost of single sensor; Passive	Stationary issue: Cannot detect motionless objects; Small range	High
Ambient (Wireless)	Radar / Wi-Fi	Wireless; Easy to install	Reliability: Very prone to interference; Range problems	High
Ambient (Floor)	Pressure mats	Accurate localization	Scalability: Extremely high cost and wiring complexity for large areas	High

Motivation for the Thesis: The analysis of the state-of-the-art reveals that there is a strong demand for a system that is privacy-preserving (like PIR), accurate in detecting stationary presence (like pressure mats) and doesn't require hundreds of units to cover large area while preserving low cost to be affordable for companies as well as private customers.

Conventional pressure mats are the closest functional equivalent to the desired solution, but their deployment cost scales with area in a way it excludes them economically. This creates an opportunity for Fiber-Optic Sensing technology.

2. Optical fiber and Fiber-Optic Sensors

2.1. Structure and Operating Principle of an Optical Fiber

2.1.1. Operating Principle

An optical fiber is a cylindrical dielectric waveguide designed to transmit electromagnetic waves in the optical spectrum (typically visible and near-infrared) [24]. In comparison to metallic conductors, that transmit electrical current, fibers guide photons through transparent medium based on varying refractive indices. A standard optical fiber consist of three concentric cylindrical layers, each performing a specific function:

- **Core:** The central physical medium where the optical signal propagates. It is typically made of high-purity silica (SiO_2) doped with elements such as Germanium or Boron to manipulate the refractive index value, marked as n_1 [24]. The diameter of the core varies significantly depending on the fiber type: typically 7-10 μm for Single-Mode Fibers (SMF) and 50 μm or 62.5 μm for Multi-Mode Fibers (MMF).
- **Cladding:** An outer optical layer surrounding the core, usually also made out of silica glass, but with slightly lower refractive index value, referred to as n_2 ($n_1 > n_2$) [24]. This difference in refractive indices is the fundamental condition required to guide light through the fiber core. The standard outer diameter of the cladding is 125 μm for both SMF and MMF fiber types.
- **Buffer coating:** Buffer surrounds core and cladding and protects it from environmental disturbances and mechanical stresses. It does not participate in light propagation but keeps the fiber reliable and safe.

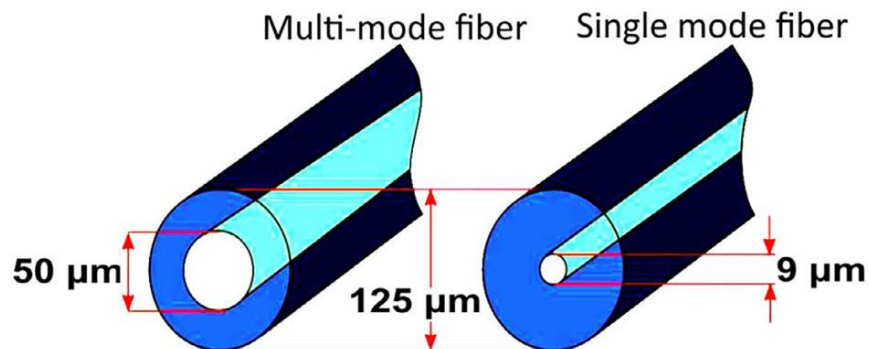


Figure 5 Cross sections of SMF and MMF fibers with dimensions [25].

2.1.2. Propagation Principle

Light traveling through the fiber obeys to the laws of geometrical optics, specifically Snell's Law and the phenomenon of Total Internal Reflection (TIR) [24]. It simply means that when a light ray is injected to the core (denser medium, n_1), it strikes the interface with the cladding (rarer medium, n_2), it is reflected back into the core of the angle of incidence θ (measured from the normal to the interface) exceeds the Critical Angle (θ_c).

From the Snell's Law [24]:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1} = n_{21} \quad (5)$$

Where:

- θ_1 – the angle between the incident ray and the normal to the surface,
- θ_2 – the angle between the reflected ray and the normal,
- n_1 – refractive index of the first medium,
- n_2 – refractive index of the second medium,
- n_{21} – relative refractive index of the second medium relative to the first.

The critical angle can be derived from the Snell's Law, by selecting the reflection angle to 90 degrees:

$$\sin\theta_c = \frac{n_2}{n_1} \quad (6)$$

Consequently, for TIR to occur, two conditions must be met:

- The refractive index of the core must be higher than that of the cladding ($n_1 > n_2$).
- The incident angle of the light ray must be greater than the critical angle ($\theta > \theta_c$)

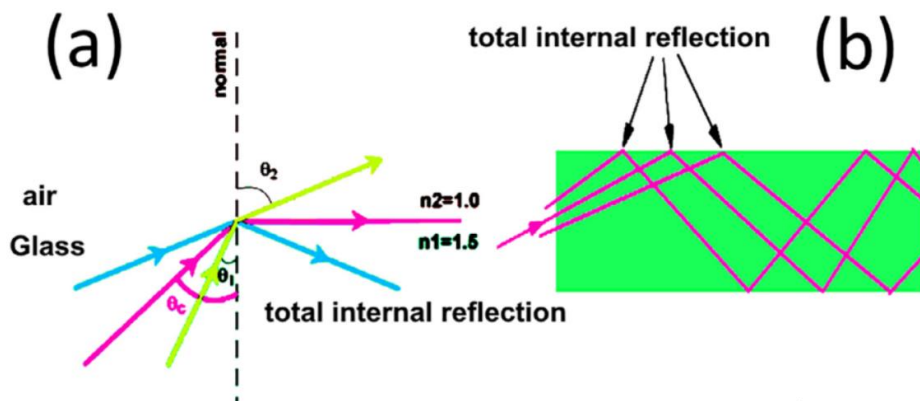


Figure 6 TIR principle, (a) in air and glass medium, (b) in fiber structure [25].

2.1.3. Numerical Aperture and Acceptance Cone

From an engineering perspective, a critical parameter is the ability of the fiber to gather light from the source (e.g. LED or laser) with the highest efficiency possible. This parameter is defined as the numerical aperture (NA). It represents the sine of the maximum half-angle (θ_a), also known as the acceptance angle, at which light can enter the fiber and undergo the TIR within the core. The NA is the dimensionless quantity related to the refractive index contrasts:

$$NA = \sin\theta_a = \sqrt{n_1^2 - n_2^2} \quad (7)$$

A higher NA causes wider acceptance cone, meaning the fiber can couple more light from incoherent sources and is more tolerant to bending losses.

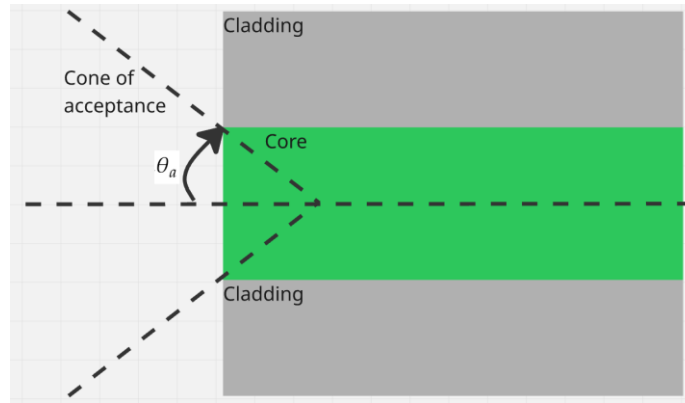


Figure 7 Acceptance cone of optical fiber. (Source: Own elaboration)

2.1.4. Fiber Classification

Fibers are categorized based on the number of electromagnetic propagation paths (modes) they support. This is mathematically determined by the **Normalized Frequency** (also known as the v-number), defined as:

$$V = \frac{2\pi a}{\lambda} \cdot NA = \frac{2\pi a}{\lambda} \sqrt{n_1^2 - n_2^2} \quad (8)$$

Where a is the core radius and λ is the wavelength of light.

- **Single-Mode Fibers (SMF):** Defined by the condition $V < 2.405$ [24, 25]. Due to small core diameter ($\approx 9\mu m$), it supports only one, fundamental mode. SMF exhibit the lowest attenuation making it the standard for long-haul telecommunication.
- **Multi-Mode Fibers (MMF):** Has a larger V-number ($V \gg 2.405$) and supports hundreds of modes. The larger core ($50/62.5\mu m$) simplifies coupling with light source and connectors.

2.2. Historical Development of Fiber Sensors

First Steps in Sensing: Fiber-optic sensing dates back to the mid-1960s, the same time of development of low-loss optical fiber telecommunication. While the primary goal of pioneers was to minimize signal attenuation to transmit data over long distances, researchers quickly observed that optical fibers were very sensitive to environmental perturbations such as temperature, strain and vibration [26].

The first commercial realization of this concept was the “Fotonic” sensor, patented in 1967, which used bifurcated fiber bundles to measure the displacement and vibration based on the intensity modulation [26]. However, the true breakthrough occurred in the mid-1970s with the focus shifting to interferometric sensors. The development of Fiber Optic Gyroscope (FOG) and high-sensitivity hydrophones for underwater acoustic detection demonstrated that optical fibers could surpass conventional electromechanical sensors in precision and immunity to EMI [26].

Microbending: In the 1980s, a new direction emerged focusing on cost-effective industrial applications. Researchers started to investigate the principle of microbending, which previously were considered as a pure loss in telecommunication. This phenomenon refers to microscopic axial deformations of the fiber, typically with the amplitudes in the micrometer range. These occur when the fiber is pressed against a rough surface by external force.

Mechanism: As described by Berthold [27], the signal loss is caused by the coupling of light from stable guided modes (traveling through the core) to unstable radiation modes (leaking into the cladding). In simple terms, the mechanical deformation changes the angle of reflection inside the fiber violating the TIR principle. This causes the light to escape from the core into the cladding resulting in a drop of optical signal intensity at the output.

Applications: Because this mechanism directly converts mechanical force into optical loss, it became the basis for industrial sensors. Microbending sensors have been successfully applied to measure variables such as pressure, strain and acceleration. In context of this thesis, this property is exploited to create a large-area pressure sensor for detecting human footstep.

There are plenty of optical fiber sensors, ranging from simple point sensors to complex distributed systems capable of monitoring kilometers of infrastructure. Table 5 presents an analysis of key fiber-optic technologies currently used in research and industry.

Table 5 Comparison of fiber-optic sensing technologies based on operating principle and complexity [26, 28-30]

Sensor Category	Technology	Operating Principle	Key Application	Complexity/Cost	Ref
Intensity - Based	Microbending	Loss of optical power due to mechanical deformation	Pressure monitoring systems	Low / Low	[27], [31], [32]
Point Sensors	FBG (Fiber Bragg Grating)	Shift in reflected wavelength caused by strain	Structural Health Monitoring	Medium / Medium	[29], [33], [34], [35]
Interferometric	Sagnac / FOG	Phase shift between counter propagating beams	Navigation (Gyroscopes)	High / High	[26], [36], [37]
Distributed (Rayleigh)	OTDR / DAS	Analysis of Rayleigh backscattering intensity/phase	Intrusion detection / acoustic sensing	High / High	[28], [38], [39]
Distributed (Brillouin)	BOTDA	Analysis of Brillouin frequency shift	Pipeline monitoring (> 50 km)	Very high / High	[28], [41]
Distributed (Raman)	RDTS	Analysis of Raman Anti-Stokes/Stokes ratio	Fire detection in tunnels	High / High	[28], [42]

The classification presented in Table 5 illustrates the versatility of optical fibers as sensing elements. While distributed technologies and interferometric sensors represent highest sensitivity and spatial resolution, their implementation relies on highly sophisticated interrogation units often costing tens of hundreds of dollars. Since the primary objective of this thesis is to develop a **cost-effective** and scalable solution for residential monitoring, a detailed analysis of these complex phenomena (Brillouin, Raman scattering, interferometry) is considered outside of the scope of this work. They are listed to provide a complete background and to contrast their complexity with the simplicity of how intensity-based microbending approach can work with.

2.3. Rayleigh Scattering and OTDR

Advanced distributed sensing systems utilize the intrinsic scattering properties of the fiber to perform precise spatial localization. The most significant technology in this field is based on Rayleigh Scattering and is commonly implemented using Optical Time-Domain Reflectometry (OTDR).

2.3.1. Mechanism of Rayleigh Scattering

Rayleigh Scattering phenomena is caused by microscopic density fluctuations in the silica glass structure, which are “frozen” during the fiber manufacturing process. These impurities are much smaller than the wavelength of the light. As the light propagates through the core, a small fraction of photons strikes these scattering centers and is deflected in all directions. Critically, a portion of this light returns to the source as backscattered light [28].

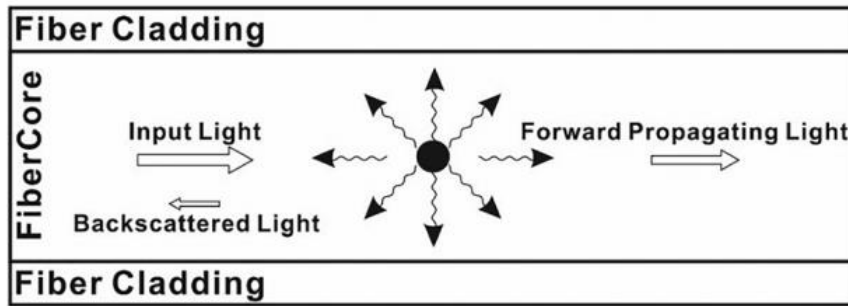


Figure 8 Schematic of spontaneous Rayleigh scattering in optical fiber [28].

Unlike microbending, which is an external loss mechanism, Rayleigh scattering occurs continuously along the entire length of the fiber. The intensity of the scattered light I_R is inversely proportional to the fourth power of the wavelength λ [28]:

$$I_R \propto \frac{1}{\lambda^4} \quad (9)$$

This relationship dictates that shorter wavelengths (e.g. visible light) are scattered significantly more than infrared signals used in telecommunication.

2.3.2. Principle of OTDR Localization

The OTDR device exploits this backscattering effect to create a virtual map of the fiber. It injects high-power optical pulses into the fiber and measures the intensity of the returning “echo” as a function of time [24]. The location of any event along the fiber is calculated using Time-of-Flight equation:

$$L = \frac{c \times t}{2n_1} \quad (10)$$

Where:

- L – the distance to the event,
- c – speed of light in vacuum ($3 \cdot 10^8$ m/s),
- t – time delay between the pulse emission and reception.
- n_1 – refractive index of the core.

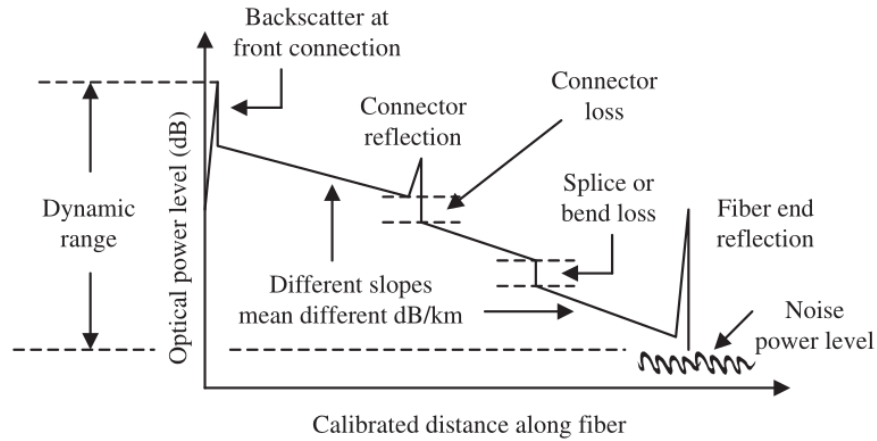


Figure 9 Representative trace of backscattered optical power displayed on an OTDR screen [24].

2.3.3. Detection of Microbending and Wavelength Trade-off

Using Rayleigh backscattering, an OTDR can detect microbending induced by a footstep. On the OTDR trace, such an event does not appear as a reflection peak, but as a localized step loss (attenuation drop) – the backscatter level after the stepping point is lower than before it, because the microbending has ejected a portion of the light into the cladding.

However, designing such a system involves a critical trade-off regarding the operating wavelength:

- **Rayleigh Advantage (Short λ):** According to the scattering law ($1/\lambda^4$), shorter wavelengths provide a stronger backscattering signal.
- **Microbending Sensitivity (Long λ):** Longer wavelengths are more prone to exhibit bending losses [24].

Therefore, while a shorter wavelength is better for observing the backscattering (via Rayleigh), a longer wavelength is superior for sensing the pressure (via Microbending).

2.4. Motivation for the Intensity-Based Approach

The analysis of state-of-the-art technologies presented in section 2.3 demonstrates that while OTDR systems offer superior spatial resolution, they suffer from a fundamental barrier to entry in residential application: hardware complexity and cost. Detecting the exact location of a microbending effect via Rayleigh backscattering requires resolving time delays in nanosecond range, pulse laser drivers and precise timing circuits. This makes reflectometric systems too expensive for a standard home installation.

To address this limitation, this thesis proposes a shift from reflectometry to measurement of the forward transmission. The core hypothesis is that the sophisticated localization provided by OTDR is unnecessary if the optical fiber is spatially distributed in a known pattern. Instead of asking “*where?*” along a single long fiber the attenuation occurred, the system divides the room into discrete zones, each monitored by specific fiber combination.

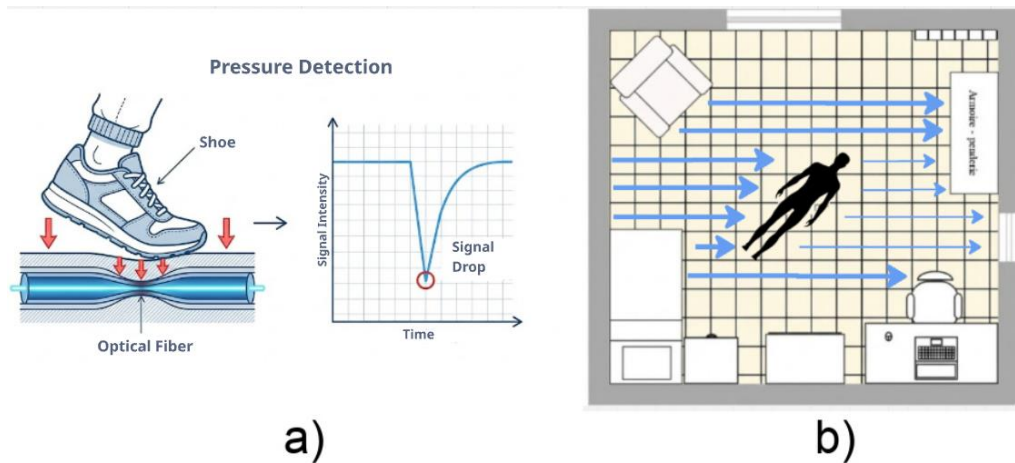


Figure 10 System operation concept: (a) Microbending effect caused by a footstep, resulting in signal attenuation; (b) Optical fiber grid layout covering the room, where blue arrows indicate the direction of signal propagation. (Source: generated using AI).

By monitoring signal intensity drop in every channel and analyzing outcome combination, the presence can be detected with the resolution equal to tile dimensions. By using constant wave light source rather than pulse and simple photodiode circuit as a receiver, the system is a fraction of cost of commercial OTDRs. This approach allows for easy scaling – by adding another fiber or changing fiber pattern – and since the light source and detectors can be placed in one device, the problem of mounting dozens of units in one room is solved.

3. Materials and methods

Design Objectives: This chapter describes the practical implementation of the fiber optic based human detection system. The primary objective was to design a cost-effective, intensity based sensor capable of detecting human footstep history in a room. The following sections describe the complete hardware and software development process, organized into three key domains:

- **Optical Setup:** Selection of the fiber type, light source, and physical sensor layout.
- **Analog Front-End:** Design of the signal conditioning circuitry to convert optical intensity changes into digital data.
- **Digital Processing:** Implementation of the data acquisition unit and the algorithmic approach for event detection.

3.1. System Architecture

High-Level Overview: The high-level architecture of the developed system is illustrated in Fig. 8. The design is built of three main functional blocks: the data acquisition and control device, the optical sensor mat, and the PC workstation for data visualization.

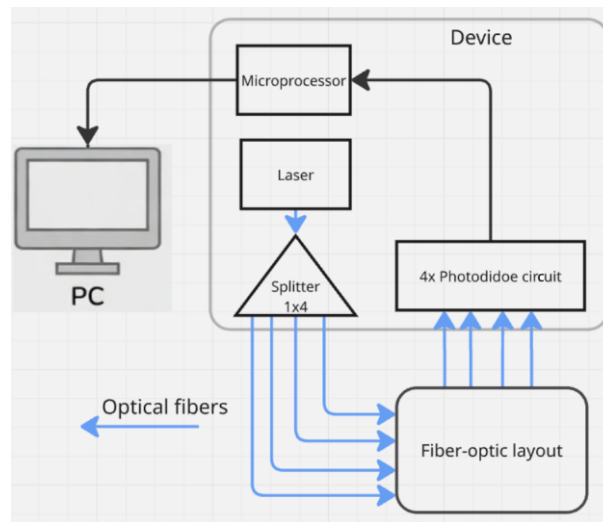


Figure 11 System architecture.

The operation principle relies on the intensity modulation of light caused by mechanical stress applied to the fiber. The signal flow proceeds as follows:

1. **Optical Transmission:** The system utilizes a transmission-mode configuration. A laser diode emits a CW optical signal, which is distributed via a 1x4 optical splitter into four independent optical fiber loops embedded in the mat.
2. **Sensing Mechanism:** The optical fiber mat serves as the intrinsic sensor. When a person steps on the mat, the mechanical deformation (microbending) causes a temporary attenuation of the optical signal in the corresponding fiber loop.

3. **Signal Acquisition:** The output of each fiber loop is coupled to a corresponding photodetector within the main device. The photodiode converts the varying optical power into an electrical current.
4. **Processing and Communication:** Each channel possesses a dedicated signal conditioning circuit connected to the microprocessor. The microcontroller digitizes the analog signals, performs initial analysis to detect events, and transmits the data to the PC via a USB interface.
5. **Visualization:** A dedicated Python-based application on the PC receives the data stream to display the current position of the subject, the history of movement, and the real-time voltage levels (corresponding to intensity) for each sensor channel.

3.2. Optical and Hardware Component Selection

3.2.1. Optical Fiber

Selection Criteria: The selection of the optical fiber was the most important decision in the hardware design process, as the fiber itself acts as the main sensing element. The material had to meet two contradictory requirements: high sensitivity to mechanical pressure (microbending) and sufficient mechanical durability to withstand repeated footsteps without fracturing.

Phase 1 - Glass Standard Selection: The primary goal of the sensor is not just binary presence detector (0/1), but also estimation of subject's weight. This helps to minimize the rate of false alarms triggered by for example pets. This requires a fiber that offers a broad dynamic range, which means that the attenuation should increase gradually with pressure applied, rather than dropping to zero immediately. Three fibers types were evaluated:

Table 6 Comparison of optical fiber standards

ITU-T Standard	Type	Core Diameter	Observed Behavior (Experimental)	Verdict
G.651.1 [43]	MMF	50 μm	Unstable: Different results with similar cases tested. Probably caused by high modal noise.	Rejected
G.652 [44]	SMF	$\sim 9 \mu m$	Binary Response: Extremely high sensitivity to microbending. Very easily saturated, preventing weigh differentiation	Rejected
G.657 [45]	SMF	$\sim 9 \mu m$	Linear Response: The "bend insensitive" trench structure prevents immediate signal loss. Allows for measure gradual drop.	Selected

Decision: The system is chosen to operate using ITU-T G.657 (Bend-Insensitive Single-Mode Fiber). While counter-intuitive for a microbending sensor, the “bend resistance” of G.657 proved to be an advantage. Standard G.652 fibers exhibited a “digital” behavior – stepping on the fiber resulted in immediate blackout (infinite attenuation). In contrast, the G.657 fiber maintained partial transmission under load, allowing to distinguish “how much” pressure was applied.

Phase 2 - Buffer Layer Analysis: During the initial testing phase, various protective coatings were evaluated. It was observed that the mechanical properties of the buffer layer significantly influence the sensor's performance:

- **Bare fiber (250 μm acrylate coating):** Proved to be too fragile for floor installation, exhibiting a high risk of breakage under standard load.
- **Standard Patch cord (3 mm jacket):** The thick jacket layers absorbed too much mechanical stress. The fiber was durable but completely insensitive to pressure.
- **Tight Buffer (900 μm):** This structure was identified as the optimal solution. The 900 μm buffer was thin enough to transfer the mechanical force of a footstep directly to the fiber core, while in the same time was thick enough to allow repeatable pressing without breakage of the fiber.

Conclusions: The final sensor element is a G.657 SMF fiber with a 900 μm Tight Buffer. This configuration assures stability and repeatable measurements.

3.2.2. Optical Transmitter Stage

3.2.2.1. Light source

Light Source Technology Selection: The selection of the optical transmitter was a multi-stage process. The selection involved: transmitter type, operating wavelength, mechanical light coupling to the fiber and driving topology.

Phase 1 – Semiconductor Technology: First, the transmitter type was considered. "The four main laser types are the Fabry-Perot (FP) laser, the distributed-feedback (DFB) laser, tunable lasers, and the vertical cavity surface-emitting laser (VCSEL)" [24], additionally LED diode was taken into account.

Table 7 Comparison of optical sources for intensity-based sensing [24].

Source Type	Spectral Width	Beam Characteristic	Coupling to SMF (9 μ m)	Cost	Verdict
LED (Light Emitting Diode)	Broad (50-100nm)	Spreads in every direction	Impossible: Because of beam characteristic, high coupling loss due to NA mismatch.	Very low	Rejected
VCSEL (Vertical Cavity)	Narrow (<1nm)	Circular beam	Difficult: Optimized for MMF, usually too low power for long transmission.	Low	Rejected
FP Laser (Fabry-Perot)	Moderate (2-5 nm)	Elliptical beam	High Efficiency: Standard for telecommunication, high output power.	Medium	Selected
DFB Laser (Distributed Feedback)	Very narrow (0.1 nm)	Elliptical beam	High Efficiency: Requires optical isolator to prevent instability from reflections.	High	Rejected
Tunable Laser	Variable	High Precision	High Efficiency: Complex packaging with integrated wave-lockers.	Very High	Rejected

Decision: LEDs were rejected immediately. It would be very hard to gather beams so they fit into the acceptance cone of the fiber, beams spread in all directions and most of the power is lost already on the input. DFB and Tunable Lasers were rejected because of their cost and complexity. The Fabry-Perot laser technology was selected for this device.

Wavelength Selection: Regarding the operating wavelength, the longer the wavelength the more it is prone to bending [24], therefore 1625nm wavelength was considered, but although being more responsive, was not chosen because of its lower availability and higher cost. Therefore 1550nm wavelength was chosen.

Phase 2 – Driving Topology: The method of supplying the laser is critical for the stability of signal and lifespan of the laser

- **Current Supply:** A laser diode driven by a Constant Current (CC) source is thermally unstable. As the junction temperature rises, the threshold current increases, causing the optical output power to drop even if the input current remains fixed [24]. Thermal runaway after some time also simply burns the laser. To prevent such an events, the systems must use Automatic Power Control (APC), this system requires the laser to have monitor diode built in and by monitoring its current generated, adjusts the supply current, keeping the output power level constant.
- **Thermal Regulation:** For a system operating continuously passive cooling might be not enough, even when APC driver is installed. Thermal fluctuations affect also the wavelength shift [24]. To ensure maximum stability and long lifespan, Thermo-Electric Cooler (TEC) is highly required.

Two lasers were considered:

Table 8 Light sources comparison.

Feature	Philips CQF58 [46]	Advanced Photonix CVLL Series [47]
Power Control	Integrated Monitor Photodiode allows constant power output	No monitor diode, requires open-loop CC driver
Temperature Stability	Integrated TEC & Thermistor	No internal cooling, wavelength drifts $\sim 0.55 \text{ nm}/^{\circ}\text{C}$
Fiber Coupling Efficiency	High: Single Mode Pigtail	Very Low: Window package, requires external coupling to the fiber
Role in Project	Ensures measurement reliability and validates scalability.	Cannot sustain continuous output needed for the sensor.

Decision: The system is chosen to operate with **Philips CQF58** laser, mostly because of its already integrated FC connector, which eliminated the problem of coupling light into the fiber, and built in monitor diode and TEC cooler which allows to work continuously even with high output power.

Future possibilities: Using such a powerful laser allow to investigate the sensor's maximum potential and scalability . This choice validates the technology at the highest level. However, for future commercial iterations, the system requirements can be relaxed. Once the optimal performance is found with the CQF58, a lower-cost laser diode (maybe even without an integrated TEC or with lower bandwidth) can be selected to reduce unit costs, provided it supports the essential APC mode.

Choosing this laser requires designing two additional circuits, APC driver circuit and TEC driver circuit, as presented in Figure 12.

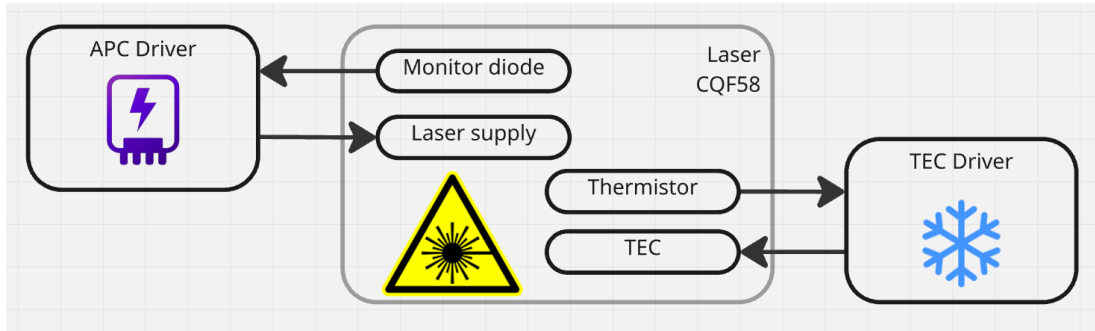


Figure 12 Circuit diagram with use of CQF58 laser.

3.2.2.2. Automatic Power Control Driver

Discrete vs. Integrated Design: Developing a custom APC circuit based on operational amplifiers to stabilize the laser power was considered. However, such designs are prone to oscillation and thermal instability if not perfectly tuned. Given the high cost of the laser diode and the risk of permanent damage due to minor design errors (e.g., current overshoot), a discrete approach was found too risky. Instead, commercially available Integrated Circuits (ICs) were evaluated. These "ready-to-use" drivers offer significant advantages, including built-in protection mechanisms such as soft-start (preventing current spikes during power-up) and fault detection (disabling the output in case of error), which are difficult to implement in discrete designs.

Driver Evaluation: Component selection and comparison of three integrated solutions were analyzed for this project. The comparison criteria focused on maximum output current, control interface complexity, and safety features. The properties of the considered candidates— LMH13000 [48], iC-WKN [49] and MAX3766 [50]—are summarized in Table 9.

Table 9 APC chips comparison.

Parameter	Maxim Integrated MAX3766 [50]	Texas Instruments LMH13000 [48]	iC-Haus iC-WKN [49]
Primary Function	Laser Driver for Fiber Comms	Precision Power Limiter Switch	CW Laser Diode Driver
Max Output Current	80 mA (Bias) / 60 mA	Adjustable (up to 5 A)	300 mA
Control Logic	PECL (high speed logic)	Logic High Enable	Analogue / Logic

Safety Features	Safety Logic, APC fault	Current Limit, Thermal Shutdown	Soft Start, Overcurrent, Overtemp
Package	TQFP	VSON-10 (Thermal Pad)	SO8
Evaluation Verdict	Rejected: Too complex (PECL) and insufficient current margin.	Rejected: Incorrect application type and difficult soldering.	Selected: Optimal current range and safety features.

- **Maxim MAX3766:** While offering nice features, this chip is designed for high-speed fiber communications. It includes PECL (Positive Emitter Coupled Logic) compatibility, which introduces unnecessary complexity for a simple CW signal application. Furthermore, the maximum bias current of 80 mA provides insufficient range for the selected laser, which may require up to 120 mA to operate at full potential.
- **Texas Instruments LMH13000:** This component is primarily a current-limited power distribution switch rather than a precision optical feedback driver. Its current range (up to several Amperes) is excessive for this application, and the VSON-10 package is difficult to solder during the prototyping phase without specialized equipment.
- **iC-Haus iC-WKN:** This driver was selected as the optimal solution. It is specifically designed for driving CW laser diodes in APC mode with currents up to 300 mA, providing a safe operating margin. Key features justifying this choice include:
 - **Soft Start:** Ensures a gradual increase of current, protecting the laser from turn-on spikes.
 - **Fault Monitor:** Automatically cuts off the power supply in case of overcurrent or overtemperature, preventing permanent damage.
 - **Simplicity:** The SO8 package and simple external circuitry allow for easy integration and soldering.

3.2.2.3. Thermoelectric Cooler Driver

Controller Strategy: Initially, specialized TEC controller ICs were considered. A representative example, the Analog Devices ADN8830 [51], was analyzed. It offers high efficiency and interesting features like PWM control and linear power stages, but it was rejected due to:

- **Cost:** Specialized TEC drivers are significantly more expensive than standard components (approx. 80 PLN vs. 33 PLN for APC drivers).
- **Complexity:** The chip requires a complex external component network and features functionality (e.g., external oscillator synchronization) that exceeds the requirements of a simple stabilization loop.

Therefore, a decision was made to design a discrete driver. The laser module is equipped with a built-in thermistor [46], allowing for direct temperature readout. The control logic is implemented using a comparator and an operational amplifier (the specific models are shared with the detection circuit and described in Section 3.2.3). The comparator controls a switching transistor that regulates current flow to the Peltier element based on the thermistor's feedback.

Switching Element Selection: The key requirement for the switching element was the ability to handle the maximum cooler current of 1A [46] and ensure fast switching to correct small thermal changes. The control logic operates at 5V, this forces the transistor's Gate Threshold Voltage (V_{GS}) to be below this level to ensure full saturation.

Two N-channel MOSFETs were evaluated:

Table 10 Comparison of MOSFET transistors for TEC Driver circuit.

Parameter	VN0606 [52]	IRFZ34N [53]	Note
Type	N-Channel MOSFET	N-Channel Power MOSFET	-
$V_{GS(th)}$	0.8V - 2.0V	2.0V - 4.0V	Both are compatible with 5V logic
$I_D(max)$	1.5A	29A	Both satisfy the max TEC current of 1A
$R_{DS(on)}$	3.0 Ω	0.04 Ω	Determines power loss
P_{max}	1W	68W	-
Power Dissipation $P = I^2R$	3W at 1A	0.04W at 1A	VN0606 would burn instantly
Verdict	Rejected	Selected	IRFZ34N operates cold without the heatsink

- **VN0606 [52]:** This transistor features a suitable threshold voltage (V_{GS} max 2.5V). However, its high On-State Resistance ($R_{DS(on)} = 3\Omega$) presents a thermal issue. At a load current of 1 A, the power dissipated by the transistor would exceed its maximum rating of 1 W, leading to overheating and failure.

- **IRFZ34N [53]:** To eliminate thermal risks, a power MOSFET was selected. With an extremely low $R_{DS(on)}$ of 0.04Ω , the power dissipation is negligible even at high currents. It can handle up to 29A and has a threshold voltage of max 4.0 V, making it compatible with the 5V logic and ideal for driving the 1A load without requiring a heatsink.

Power Stage: For the first project iteration, the adjustable voltage regulator was considered, mainly to be able to investigate the influence of voltage to current output. A linear regulator (LDO) approach was rejected due to the high component count (external capacitors, potentiometers). Instead, a step-down converter module based on the MP1584 [54] IC was selected. It is a high-frequency switching regulator capable of delivering up to 3A of output current. This provides a significant safety margin relative to the 1A requirement, ensuring stable operation without thermal stress on the power supply unit.

3.2.3. Analog Receiver Stage

Stage Overview: Signal receiving circuit is as important as the light emitting one. In this application, the photodiode serves as the signal receiver. This signal is converted by TIA circuit and then treated with comparator circuit. Schematic can be seen in Figure 13.

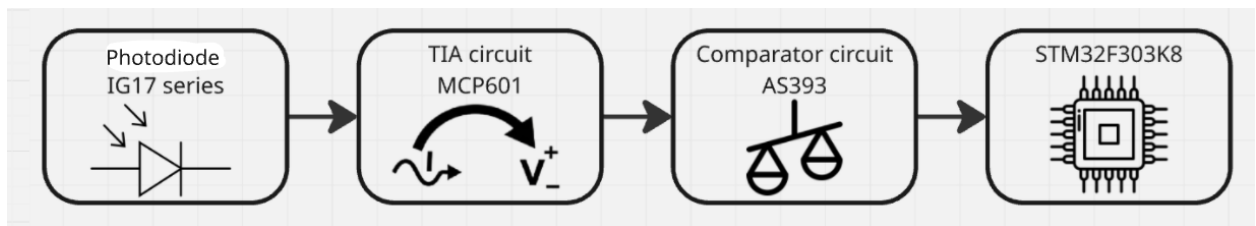


Figure 13 Detector circuit block diagram.

3.2.3.1. Photodiode

Operational Mode: The photodiode functions as the Optical-to-Electrical (O/E) converter. Operating in the photoconductive mode (reverse bias), it generates a photocurrent linearly proportional to the incident optical power [24]. In the context of this application, the light intensity correlates directly with the weight applied to the sensor; therefore, the detector must exhibit high responsiveness and stability.

Material Selection: Similarly to the laser, there are few kinds of photodiodes, they differ by material they are built from. As decided in previous section, the operating wavelength is 1550nm, therefore few photodiode types were considered.

Table 11 Comparison of photodiode material for infrared sensing [24].

Material	Wavelength Range	Peak Responsivity	Dark Current	Verdict
Silicon (Si)	400 - 1100 nm	0.4 - 0.6 A/W	1 - 10 nA	Rejected: Too narrow range
Germanium (Ge)	800 - 1650 nm	0.4 - 0.5 A/W	50 - 500 nA	Rejected: Poor responsivity and high dark current
InGaAs	1100 - 1700 nm	0.75 - 0.95 A/W	0.5 - 2.0 nA	Selected

Decision: InGaAs diode was selected for this project for its great responsivity, this parameter was twice as high as the germanium's responsivity. Additionally, germanium diodes suffer from significantly higher dark current, with one detector this value might be ignored but with hundreds of detectors for large-area applications this might be significant. Silicon diode was rejected because of its narrow wavelength range.

Component Selection: Two InGaAs photodiodes were evaluated for the receiver stage: the surface-mount SML Series [55] and the through-hole IGL Series [56]. Their key parameters are compared in Table 12.

Table 12 Photodiode models comparison.

Parameter	SML-210 Series [55]	IGL-3000 Series [56]
Package Type	Surface Mount (SMD)	Metal Can (TO-46)
Active Area	Large (1.27 × 1.25 mm)	Small (Ø 0.25 mm)
Spectral Range	1550 nm	1550 nm
Mounting	Reflow Soldering	Through-Hole
Selection Verdict	Rejected due to mechanical fragility.	Selected for robust mounting.

Packaging Selection: Comparing the two candidates, the SML series initially appeared as a better selection because of its significantly larger active detection area (1.58 mm² vs 0.05 mm² for the IGL). A larger area simplifies the optical coupling process, making the alignment between the fiber core and the detector less problematic. However, the IGL Series (specifically the 0.25mm variant) was selected due to its packaging. The hermetic TO-46 metal can package offers superior mechanical properties compared to the small SMD ceramic package of the SML diode. The TO-46 form factor allows for the design of a custom 3D-printed cap that fits over

the metal can, allowing the fiber illuminating the active area directly. In contrast, mounting a fiber directly over a microscopic SMD component on a PCB creates a difficult connection to be done.

3.2.3.2. Transimpedance Amplifier

Amplification Principle: Since the photodiode acts as a small current source and the microcontroller's Analog-to-Digital Converter (ADC) requires a voltage input, a conversion stage is necessary. A Transimpedance Amplifier (TIA) was utilized for this purpose. This circuit produces an output voltage linearly proportional to the input photocurrent, determined by the feedback resistor [57].

Design Constraints: Requirements for the ease of assembly and to minimize the PCB footprint - a dual-package amplifier (containing two op-amps in one chip) was desired. The key technical requirements included:

- **Low Power Consumption:** To ensure energy efficiency for the scalable system.
- **High Input Impedance:** Important for the usage in TEC driver circuit.
- **Rail-to-Rail Operation:** To maximize the dynamic range of the sensor utilized by the ADC.

Three common operational amplifier families were evaluated: CMOS-based MCP602 (Microchip) [58], LM358B (Texas Instruments) [59] and the general-purpose RC4558 (Texas Instruments) [60]. Their electrical characteristics are compared in Table 13.

Table 13 Operational amplifier models comparison.

Parameter	Texas Instruments RC4558 [60]	Texas Instruments LM358B [59]	Microchip MCP602 [58]
Technology	Bipolar	Bipolar	CMOS
Rail-to-Rail Output	No	No ($V \pm 1.5 V$)	Yes
Quiescent Current (I_q) per channel	2.5 mA (typ)	300 – 500 μA	230 μA (typ)
Input Bias Current (I_B)	80 nA	10 nA	1 pA
Input Impedance	550 M Ω	Not specified (High)	$10^{13} \Omega$
Bandwidth (GBWP)	4 MHz	1.2 MHz	2.8 MHz

The **RC4558** was rejected primarily due to its power consumption. With a typical supply current of 2.5 mA, it consumes significantly more energy than modern alternatives. Furthermore, its input resistance is limited to approximately 2 M Ω , which is less ideal for high-sensitivity TIA applications compared to CMOS inputs.

The **LM358B** suffers from a critical limitation regarding the output voltage swing. It is not a Rail-to-Rail device; its maximum output voltage is limited to $V_{CC} - 1.5V$. In a 3.3V logic system, this would cap the sensor reading .

The **MCP602** was selected as the superior solution. As a CMOS device, it features an ultra-low input bias current of 1 pA, effectively offering infinite input impedance for this application. Its supply current satisfies the low-power requirement. Most importantly, it features Rail-to-Rail Output, allowing the signal to swing from 0V to V_{DD} , utilizing the full resolution of the microcontroller's ADC.

3.2.3.3. Comparator

Dedicated IC Rationale: Although operational amplifiers can function as comparators when used in positive feedback configuration, dedicated comparator ICs are better optimized for this specific purpose [57]. Dedicated comparators are designed to prevent the saturation delays typical of op-amps, which results in faster response times and clean digital output switching compatible with logic levels.

The primary objective was the same as with choosing the TIA OpAmp, to find a dual-channel comparator (to minimize component count) with the lowest possible power consumption to maintain the system's energy efficiency.

Candidate Evaluation: Two pin-compatible, industry-standard dual comparators were evaluated: the Texas Instruments LM393B [61] and the Diodes Incorporated AS393 [62]. Their key electrical specifications are compared in Table 14.

Table 14 Comparator chips comparison.

Parameter	Texas Instruments LM393B [61]	Diodes Inc. AS393 [62]
Supply Current (I_{CC})	0.4 mA (Typ) / 1 mA (Max)	0.6 mA (Typ) / 1.0 mA (Max)
Input Offset Voltage (V_{IO})	1.0 mV	1.0 mV
Output Sink Current	16 mA	16 mA
Operating Voltage Range	2 V to 36 V	2 V to 36 V
Cost	3PLN per unit	1PLN per unit
Verdict	Rejected: Marginally better power, but higher cost.	Selected: Most cost-effective solution.

Performance Analysis: Both components exhibit very similar performance characteristics. The Texas Instruments LM393B offers a slight advantage in power efficiency, with a typical supply current of 0.4 mA compared to 0.6 mA for the AS393. However, in the context of the total system power budget (dominated by the laser diode), this difference (approx. 200 μ A) is negligible. Consequently, the Diodes Inc. AS393 was selected. It meets all technical requirements regarding speed and logic compatibility while being the more economically viable option.

3.2.4. Processing Unit

Central Role: The processing unit serves as the heart of the device, responsible for coordinating three critical tasks: data acquisition from the analog front-end, digital signal processing (event detection), and communication with the user interface.

PCB Complexity Challenges: Using a high-performance microcontroller capable of USB and potential future Ethernet communication requires a complex PCB design. Key challenges include impedance matching for differential pairs, multi-layer stack-ups, and precise soldering of small-pitch SMD components.

Hardware Strategy: To accelerate the prototyping phase and minimize hardware risks, a decision was made to utilize a commercial-off-the-shelf (COTS) Evaluation Board rather than designing a custom circuit around a bare microcontroller.

Selection Requirements: The selection process was conducted by the following system requirements:

- **I/O Count:** Support for at least 8 analog inputs (ADC and COMP channels) for the sensor array, with a possibility to expand.
- **Communication:** Native USB interface for real-time data streaming and alarming.
- **Scalability:** Potential for Ethernet connectivity (for local network integration) and additional interfaces (I2C/SPI) for possible future sensor expansion.
- **Form Factor:** The device must remain compact.

Three development platforms were evaluated: the high-performance STM32 Nucleo-144 (F767ZI) [63], the wireless-centric ESP32-S3-DevKitC [64], and the compact STM32 Nucleo-32 (F303K8) [65]. Their characteristics are summarized in Table 15.

Table 15 Microprocessor evaluation boards comparison.

Parameter	Nucleo-144 (F767ZI) [63]	ESP32-S3-DevKitC [64]	Nucleo-32 (F303K8) [65,68]
Core Architecture	ARM Cortex-M7 (216 MHz)	Xtensa® Dual-Core 32-bit	ARM Cortex-M4 (72 MHz)
Form Factor	Large (ST Zio / Morpho)	Compact (Arduino Nano)	Compact (Arduino Nano)
Connectivity	Ethernet, USB OTG	WiFi, Bluetooth LE, USB OTG	USB 2.0 FS
I/O Flexibility	High (144 pins)	Excellent (Pin MUX)	Moderate (32 pins)
ADC Units	3× (up to 24 channels)	2× (SAR ADC)	2× (High Speed)
Verdict	Rejected: Oversized physical footprint.	Rejected: Wireless features not required.	Selected: Optimal balance of size and functionality.

Platform Justification:

1. **STM32 Nucleo-F767ZI:** While offering superior performance and a built-in Ethernet interface compliant with IEEE-802.3-2002 [63], this board was rejected due to its physical size. The Nucleo-144 form factor is too large for the intended compact dimensions of the final device.

2. **ESP32-S3:** This platform offers a significant advantage in terms of routing flexibility; thanks to its IO MUX [64], any function can be mapped to any physical pin, simplifying PCB layout. However, its primary features (WiFi/Bluetooth) are not critical for this stage of the project.
3. **STM32 Nucleo-F303K8:** Ultimately, the F303K8 was selected. Although it has static pin definitions (lacking the full flexibility of the ESP32 matrix), it provides a sufficient number of Analog I/O pins [65]. The decisive factor was the developer's familiarity with the STM32 ecosystem. Using a known architecture minimizes software development time and reduces the risk of configuration errors.

3.2.5. Power Management

Operational Necessity: Stable power delivery is fundamental for the device's continuous operation. Since the system is designed for stationary room monitoring (potentially 24/7), battery power was rejected because of the need for frequent replacement or recharging. Instead, the system is powered by the mains supply.

Main Power Source: The system is supplied by an external 12V DC power adapter. This voltage was chosen to integrate with the high-power components:

- **Microcontroller:** The STM32 Nucleo-32 board accepts an input voltage (V_{IN}) range of 7-12 [65], allowing for direct connection to the main bus.
- **Thermal Control:** The MP1584 [54] step-down converter for the TEC cooler accepts 4.5-28V input range.

Regulation Strategy: To supply the laser, logic and analog sensing circuits, two linear regulation stages were implemented. A linear approach (LDO) was chosen over switching regulators for these lines to minimize switching noise, which is critical for the precision of the analog optical measurements.

5V Domain Analysis: The laser logic and the APC driver operate at 5V. A power budget analysis was conducted:

- **APC Driver:** Draws a maximum total current ($I_{total} = I_{DC} + I_{ldk}$) of maximum 140 mA (20 mA bias + 120 mA laser drive)
- **Op-Amps:** The TIA and signal conditioning stages consume up to 30 mA under worst-case output loading
- **Comparators:** Estimated peak consumption of approx. 50 mA.

The total estimated load is approximately 220 mA. To provide a safety margin and ensure thermal stability, the LF50C Low Drop-Out (LDO) regulator was selected. It offers a maximum output current of 500 mA, providing a (>50%) safety margin. Furthermore, the LF50C was chosen for its TO-220 package, which allows efficient heat dissipation and easy soldering.

3.3V Domain Analysis: The microcontroller logic and the photodiode detection circuit operate at 3.3 V. Direct connection to 5 V logic could permanently damage the STM32 I/O pins.

For this stage, the Microchip MCP1825 was selected. Key reasons for this choice include:

- **High Current Capability:** Like the LF50C, it supports up to 500 mA, allowing for the future addition of multiple peripheral sensors without redesigning the power section.
- **Low Noise:** It offers high Power Supply Ripple Rejection (PSRR) [67], ensuring a clean voltage reference for the ADC.

Cascaded Topology: A specific design constraint of the MCP1825 is its maximum input voltage of 6.0 V. Consequently, the system utilizes a cascaded topology: the 12 V main line is first regulated down to 5 V by the LF50C, which then feeds the MCP1825 to generate 3.3 V.

The thermal load is split between two devices, improving the overall thermal reliability of the PCB.

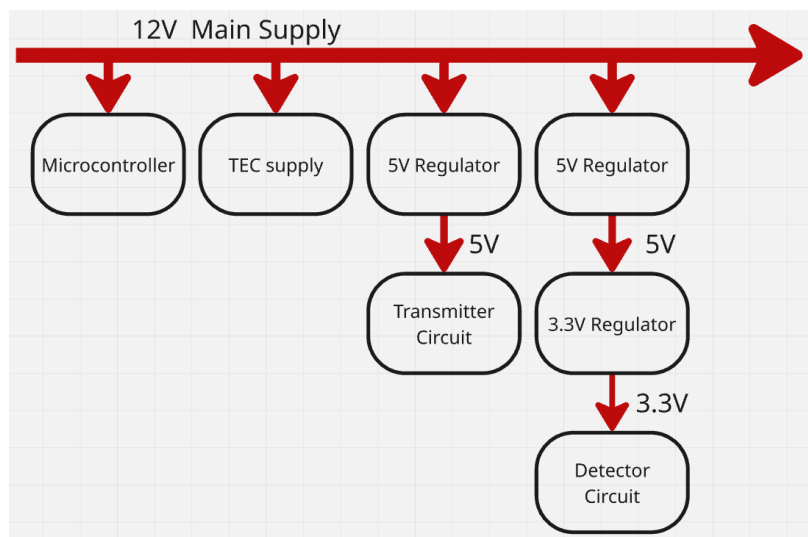


Figure 14 Power supply topology.

3.3. Optical Fiber Layout and Placement

Resolution Challenge: The physical arrangement of the optical fibers determines the spatial resolution of the system. Given the hardware limitation of four independent optical channels (as described in Section 3.1), a standard one-to-one mapping would limit the detection to only four large zones. To overcome this, a specialized topological strategy was developed. The objective was to map the room into distinct detection zones (pixels).

3.3.1. Topology Strategies

Three layout strategies were mathematically analyzed to maximize the resolution, results can be seen in Table 16.

Table 16 Optical fiber layouts comparison.

Strategy	Linear Strips	Cartesian Grid	Binary-Coded Matrix
Geometry	4 Parallel Lines	2 Vertical × 2 Horizontal	Overlapping Zones
Mathematical Limit	n	$(n/2) \times (n/2)$	$2^n - 1$
Resolution (n=4)	4 Zones	4 Zones	15 Zones
Advantages	Simplest implementation; no fiber crossing.	Creates (X, Y) coordinates; intuitive mapping.	Maximizes hardware efficiency; highest resolution.
Disadvantages	Large blind spots; extremely low resolution.	Limited intersection points; resolution scales poorly.	Complex fiber routing; requires "truth table" decoding.

Selected Topology: The Binary-Coded Matrix approach was selected as the optimal solution. This novel topology utilizes the concept of binary logic to multiply the number of detection zones without adding extra electronic components.

By intentionally overlapping specific fibers in specific physical areas, each zone is assigned a unique 4-bit address. When a person stands in a specific zone, they trigger a unique combination of sensors. The system functions as a mechanical Look-Up Table (LUT), where the output vector corresponds to a specific location in the room. The logic of the mat sensing output is illustrated in Table 17 and Figure 15.

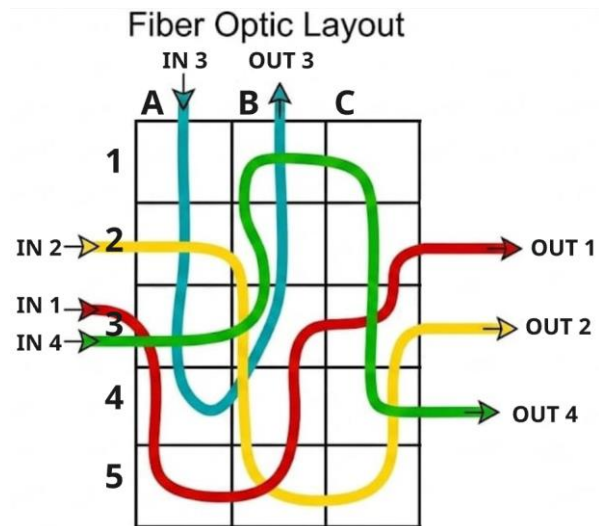


Figure 15 Optical fiber layout.

Table 17 Binary representation of comparator inputs combination.

PD1	PD2	PD 3	PD 4	Zone
0	0	0	0	X
0	0	0	1	C1
0	0	1	0	A1
0	0	1	1	B1
0	1	0	0	C5
0	1	0	1	C4
0	1	1	0	A2
0	1	1	1	B2
1	0	0	0	A5
1	0	0	1	C2
1	0	1	0	A4
1	0	1	1	A3
1	1	0	0	B5
1	1	0	1	C3
1	1	1	0	B4
1	1	1	1	B3

Complexity Trade-off: While this topology offers the highest theoretical resolution ($2^4 - 1 = 15$ zones), it introduces significant complexity in the manufacturing process. The routing requires precise placement of fibers to create specific intersection areas, unlike the simple straight lines of the linear topology. However, this trade-off is justified by the nearly 400% increase in resolution compared to standard methods.

3.3.2. Physical Construction

Mat Dimensions: Based on the binary topology, the sensing mat was constructed to cover an active area divided into 15 square zones, each with dimensions of 25 cm x 25 cm.

Substrate Evolution: The fabrication process evolved through two stages:

1. **Prototyping Phase (Laminate Underlay):** Initial tests were conducted using standard foam underlay for floor panels. This allowed for rapid verification of the binary overlap concept. However, the foam surface offered poor adhesion; the fibers tended to detach and shift positions, leading to calibration errors.
2. **Final Construction (Rubber Mat):** To achieve mechanical stability, the substrate was switched to industrial rubber mat (dimensions: 1.4 m X 1.0 m, thickness: 3mm). The rubber material provides:
 - **Stability:** A high-friction surface that prevents the mat from sliding on the floor.
 - **Elasticity:** Rubber acts as a restoring spring, improving the dynamic response of the sensor.
 - **Adhesion:** The fibers were permanently fixed to the rubber substrate using duct tape to maintain the precise geometry of the binary overlaps.

3.4. PCB Design

Motivation for PCB: The transition from the prototyping phase (breadboard) to a dedicated Printed Circuit Board (PCB) was required to ensure signal integrity and mechanical stability. Prototyping boards often introduce parasitic capacitance and inductance, leading to noise which is unacceptable for precise optical measurements. Furthermore, a PCB eliminates the risk of loose wire connections by replacing them with fixed copper signal paths. The PCB design was developed using KiCad software, while the analog circuit simulations were performed in LTSpice to verify the component values before manufacturing.

3.4.1. Transmitter circuit

Module Overview: The transmitter stage is responsible for maintaining the stability of the laser diode. This module consists of two sub-circuits: the TEC driver for temperature stabilization and the APC driver for optical power control.

3.4.1.1. TEC cooler circuit

Feedback Loop: The temperature control system operates as a feedback loop based on a Negative Temperature Coefficient (NTC) thermistor built into the laser module. The thermistor has a nominal resistance of $R_{th} = 10k\Omega$ at 25°C [46].

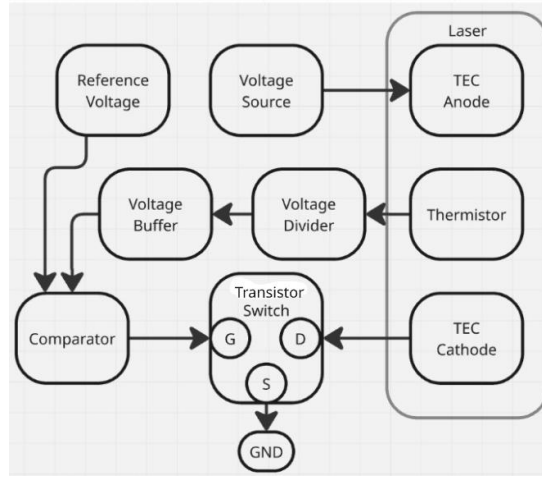


Figure 16 TEC driver topology.

Sensing Stage: To convert the resistance change into a measurable voltage signal, a voltage divider was implemented. The thermistor is placed in series with a fixed resistor $R_{fix} = 10\text{ k}\Omega$. Given the supply voltage $V_{CC} = 5\text{V}$, the output voltage V_{sense} at the target temperature of 25°C is calculated as:

$$V_{sense} = V_{CC} \frac{R_{th}}{R_{th} + R_{fix}} = 5\text{V} \frac{10k\Omega}{10k\Omega + 10k\Omega} = 2.5\text{V} \quad (11)$$

Impedance Buffering: An important feature of the laser, that needs to be remembered, is the internal interconnection between the thermistor node and the laser diode anode [46]. Direct connection to the sensing circuit could result in the "loading effect," where the measurement circuit draws current from the laser supply, or conversely, the laser current influences the temperature reading. To isolate these stages, a Voltage Buffer (Voltage Follower) based on an operational amplifier was utilized. Since an ideal op-amp has infinite input impedance [57] (in reality, the chosen CMOS op-amp draws only picoamperes of bias current [58]), it monitors the voltage divider without drawing significant current, effectively separating the sensing logic from the laser power path.

Control Logic: The buffered signal is fed into a comparator. The second input of the comparator is connected to a potentiometer, which sets the reference voltage (V_{ref}) corresponding to the desired temperature setpoint.

The comparator output controls the gate of the switching power MOSFET (IRFZ34N). Since the comparator has an Open-Drain output, a pull-up resistor is required to pull the gate high when the output transistor is on [62]. When the laser temperature rises, the thermistor resistance drops, causing V_{sense} to change. When the threshold is crossed, the comparator switches the MOSFET on, completing the circuit for the Peltier element to cool down the laser.

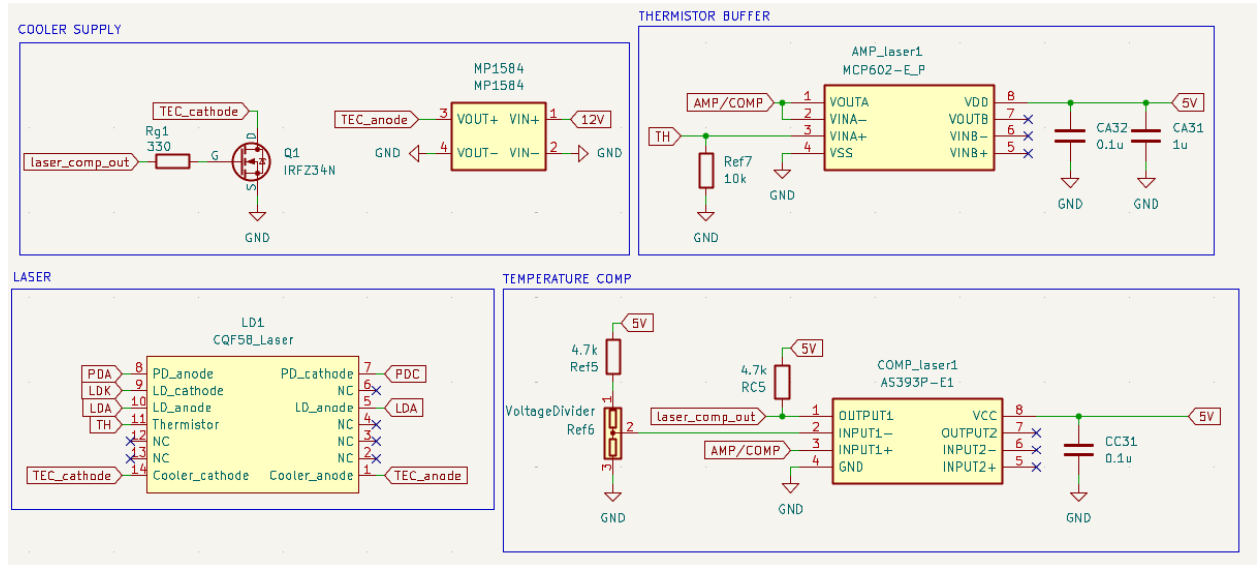


Figure 17 TEC driver electric diagram.

3.4.1.2. APC driver circuit

Implementation Details The Automatic Power Control stage is built around the iC-WKN driver IC, selected for its ability to maintain constant optical output despite temperature variations. To ensure stable operation and filter out power supply noise, the chip is powered from the 5V rail with a 100 nF ceramic decoupling capacitor placed close to the VCC pin.

The laser diode is connected in the "N-type" configuration, with the anode driven by the chip's LDA pin and the cathode connected to the LDK driver output. A critical component in this design is the capacitor connected to the CI pin. It defines the time constant of the internal control loop and enables the Soft-Start function [49]. A value of 100 nF was selected to ensure a gradual current ramp-up (typically within 70–150 μ s [49]), protecting the sensitive laser structure from destructive current spikes during the power-on sequence.

Calibration and Power Setting The iC-WKN regulates the laser current by maintaining a fixed reference voltage of 0.5 V at the monitor input (MDA pin). The optical power level is set by an external resistance (R_M) connected between MDA and AGND, which determines the target monitor diode current (I_{MD}) according to the formula:

$$R_M = \frac{V_{ref}}{I_{MD}} = \frac{0.5V}{I_{MD}} \quad (12)$$

The relationship between the laser drive current (I_{LD}) and the monitor photocurrent (I_{MD}) was investigated, an empirical characterization was performed using a laboratory power supply. The measurement results for the selected Philips CQF58 module are presented in Table 18.

Table 18 Monitor Diode current measurements.

Laser Drive Current (I_{LD})	Monitor Diode Current (I_{MD})	State
0 mA	0 μ A	OFF
10 mA	2 μ A	Below Threshold
20 mA	60 μA	Lasing Threshold
30 mA	280 μ A	Linear Region
40 mA	380 μ A	Linear Region
50 mA	490 μA	Target Operating Point

Based on the measurements, to achieve a target drive current of 50 mA (providing sufficient optical power for the sensor mat), the required monitor current is approximately 490 μ A. Substituting this into the control equation:

$$R_M = \frac{0.5V}{490\mu A} \approx 1020\Omega \quad (13)$$

To allow for the ability to dim or disable the laser, a series combination of a fixed resistor and a potentiometer was implemented.

- **Fixed Resistor (1 k Ω):** Sets the maximum current limit. With the potentiometer set to 0 Ω , the total resistance is 1 k Ω , resulting in I_{MD} =500 μ A (approx. 50 mA drive current). This acts as a hardware safety limit to prevent burning out the laser.

-

2. **Amplification:** The photocurrent (I_{ph}) is converted to voltage via a feedback resistor (R_f). A high value of $R_f = 1M\Omega$ was selected to achieve a transimpedance gain of 10^6 V/A (mapping 1pA of current change to 1μ V of voltage swing).
3. **Transfer Characteristic:** The output voltage follows the equation:

$$V_{out} = V_{ref} - (I_{ph} \times R_f) \quad (15)$$

Under normal conditions (no pressure), light intensity is high, I_{ph} is significant, and V_{out} is pulled low. When pressure is applied to the fiber (microbending), light intensity drops, I_{ph} decreases, and V_{out} rises towards the 2V reference.

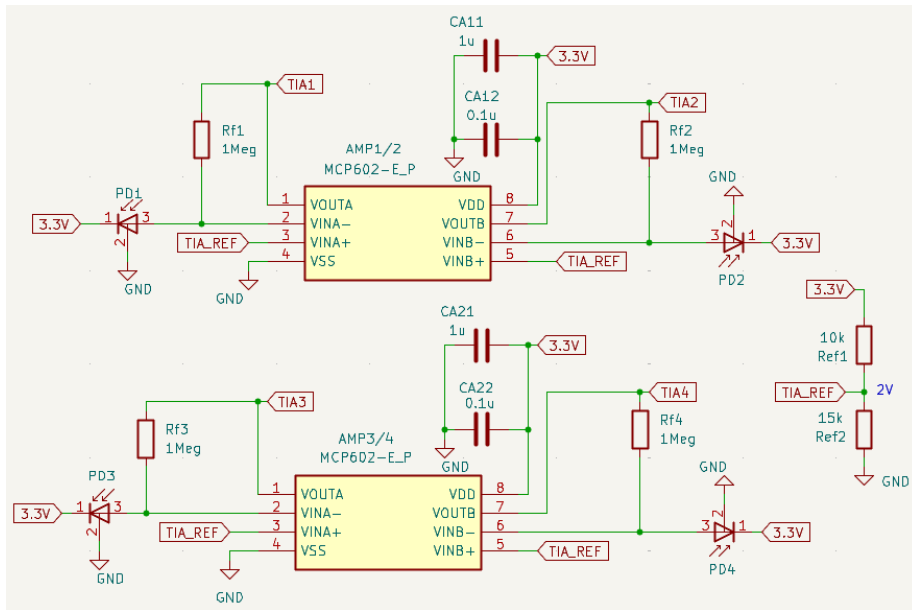


Figure 19 TIA circuits electrical diagram.

Mechanical Integration: To connect the optical fiber and the photodiode, a custom optical coupling adapter was manufactured using 3D printing technology. The element was designed to fit tightly over the TO-46 metal can of the photodiode, ensuring distinct advantages:

- **Centric Alignment:** The internal channel of the adapter guides the optical fiber ferrule directly to the center of the photodiode active area, maximizing the optical power transfer.
- **Ambient Light Rejection:** The adapter was printed using black material, which acts as a shield, preventing external light sources from affecting the sensitive measurement. Additionally, it keeps the dangerous to human eye beams inside the adapter.
- **Mechanical Stability:** It holds the fiber in place, preventing signal fluctuations caused by micro-movements of the cable relative to the detector surface.

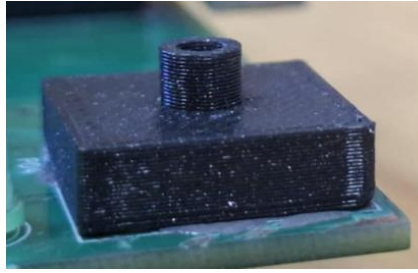


Figure 20 Fiber coupling adapter.

3.4.2.2. Comparator

Signal Routing: The analog output from the TIA is routed to two destinations: the ADC input of the microcontroller (for advanced processing) and the non-inverting input of the hardware comparator AS393.

The inverting input of the comparator is connected to a potentiometer. This allows for manual calibration of the triggering threshold (sensitivity). If the fiber is pressed, the TIA voltage rises; once it exceeds the potentiometer voltage, the comparator output flips to High.

Visual Feedback: The comparator output drives a status LED. This feature allows the device to function as a standalone unit (providing visual feedback on footsteps) even without the microcontroller or PC software, which significantly helped during the prototyping and fiber layout testing phases.

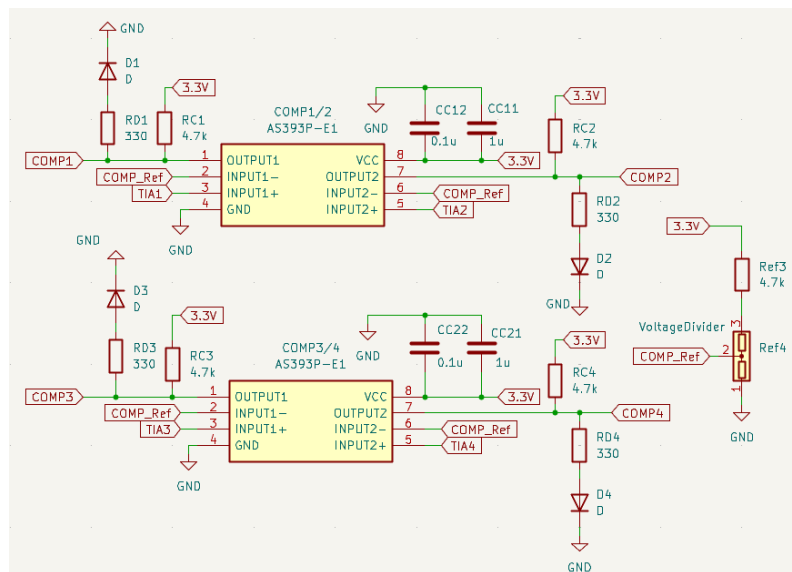


Figure 21 Comparators circuit electric diagram.

3.4.3. Microcontroller Integration and Resource Allocation

Pinout Strategy: The integration of the NUCLEO-F303K8 microcontroller with the analog front-end required a strategic assignment of the General Purpose Input/Output (GPIO) pins. The pinout configuration was designed to maximize data acquisition performance while preserving key communication interfaces for future system expansion.

Digital Interface: The binary outputs from the four hardware comparators are connected to the digital pins of the Nucleo-32 board [65]. While the current firmware iteration uses simple polling mechanism (checking the pin state in the main loop), the hardware connections were mapped to External Interrupt (EXTI) capable lines (EXTI0, EXTI1, EXTI6, EXTI7). This architectural decision ensures that the system is "future-proof." It allows for a transition to an asynchronous, interrupt-driven event detection system in future development phases without any modifications to the PCB or pinout.

Analog Interface: The analog voltage signals from the Transimpedance Amplifiers are routed to the microcontroller's ADC inputs. The STM32F303K8 features two independent ADC peripherals (ADC1 and ADC2)[65]. Instead of utilizing a single ADC for all four channels, the load was distributed symmetrically between ADC1 and ADC2 (2 channels per peripheral). This architecture provides two benefits:

1. Utilizing two ADCs in parallel reduces the total conversion time for the sensor array.
2. The pin selection intentionally avoided pins associated with hardware SPI and I2C interfaces. This reservation allows for the seamless addition of Ethernet modules in future iterations without requiring PCB rerouting.

Table 19 Nucleo board pinout along with STM32 chip pinout and functional description.

Nucleo Pin	STM32 Pin	Signal Name	Peripheral Function	Description
D3	PB0	COMP4	GPIO_EXTI0	Digital Trigger
D6	PB1	COMP1	GPIO_EXTI1	Digital Trigger
D5	PB6	COMP2	GPIO_EXTI6	Digital Trigger
D4	PB7	COMP3	GPIO_EXTI7	Digital Trigger
A0	PA0	TIA3	ADC1_IN1	Analog Level
A1	PA1	TIA4	ADC1_IN2	Analog Level
A3	PA4	TIA2	ADC2_IN1	Analog Level
A6	PA7	TIA1	ADC2_IN4	Analog Level

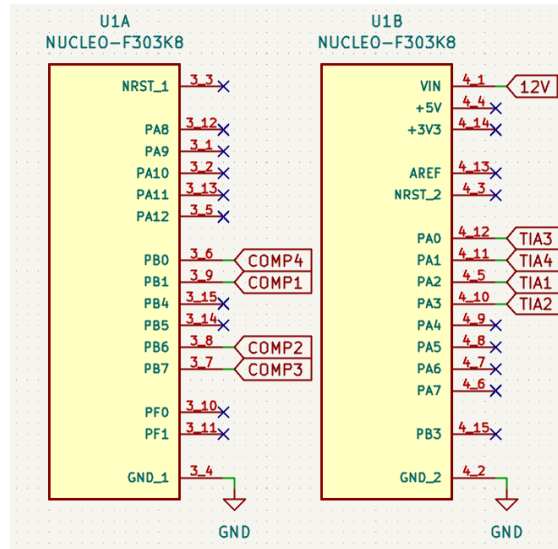


Figure 22 Nucleo board electric diagram.

3.4.4. Power Supply Implementation

The power distribution system is designed to handle mixed-signal requirements, separating high-power loads from sensitive analog measurement circuits. The system is powered by a central 12V DC bus.

The main 12V input is routed directly to two high-demand subsystems:

- **Microcontroller Board:** The STM32 Nucleo-32 is powered via its V_{IN} pin (accepting 7–12 V), utilizing its on-board regulator to power the MCU core [65].
- **Thermal Control (TEC):** The Step-Down converter (MP1584 [54]) is supplied directly from the 12 V rail to maximize efficiency when driving the Peltier element.

For the custom electronics, a linear regulation scheme was implemented to ensure low noise.

- **5V Stage (Laser Supply):** The laser diode and APC driver require a stable 5 V supply [49]. This is provided by the LF50CV LDO regulator [66]. A 100nF ceramic capacitor is placed at the input of the LF50CV (close to the V_{IN} pin). filters high-frequency noise from the main power supply.
- **3.3V Stage (Analog Front-End):** The detection circuit (Photodiodes, TIA, Comparators) operates at 3.3 V. The MCP1825 regulator was selected for this purpose. As mentioned earlier, cascaded topology was applied, on the input of every LDO, the decoupling capacitor was placed.

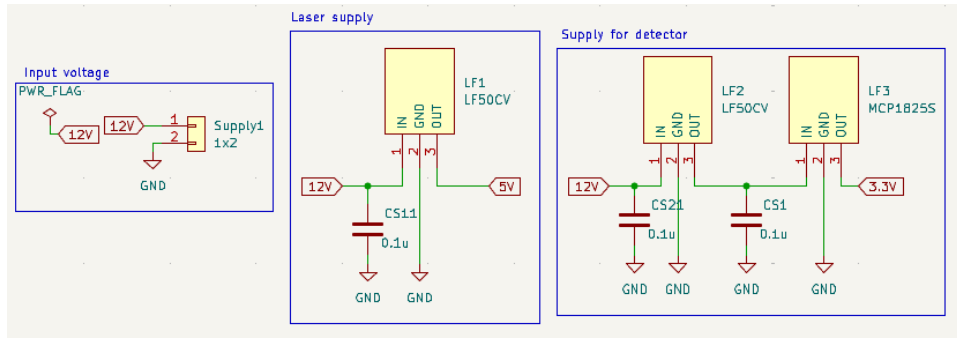


Figure 23 Power supply electric diagram.

3.4.5. Final Device Assembly

PCB Manufacturing: Following the electrical design phase, the final double-layer PCB layout was developed using KiCad software, as shown in Figure 24. Special attention was paid to the spacing between components to make the voltages and currents measurements easier.

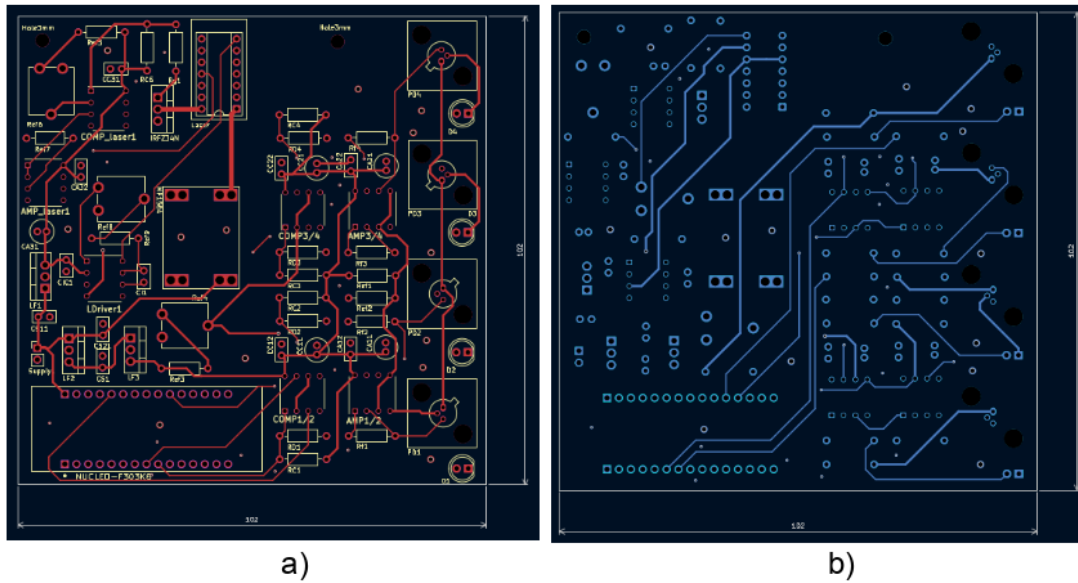


Figure 24 PCB layout. a) top side. b) bottom side.(Ground zones deleted from view)

The manufactured and soldered board is presented in Figure 25. The physical realization confirms the compact form factor assumptions, fitting all sub-circuits (Laser Driver, TEC Controller, Receiver Array, and Power Management) within the intended dimensions.

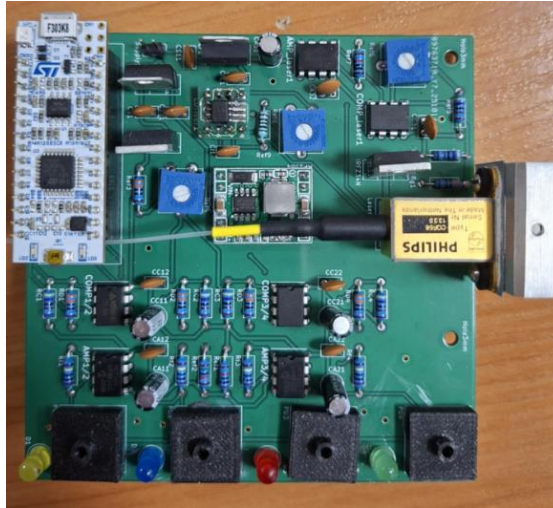


Figure 25 Manufactured and soldered PCB.

Mechanical Enclosure: To protect the device from environmental factors and provide mechanical stability for the optical connections, a dedicated housing was manufactured using 3D printing technology

The enclosure was designed to meet the following functional requirements:

- **Port Accessibility:** It features precise cutouts for the 12V power supply input, the USB data interface for PC communication, four optical fiber outputs and inputs.
- **Safety:** It isolates the user from live electrical components and protects the fragile optical couplings from accidental mechanical stress during operation.

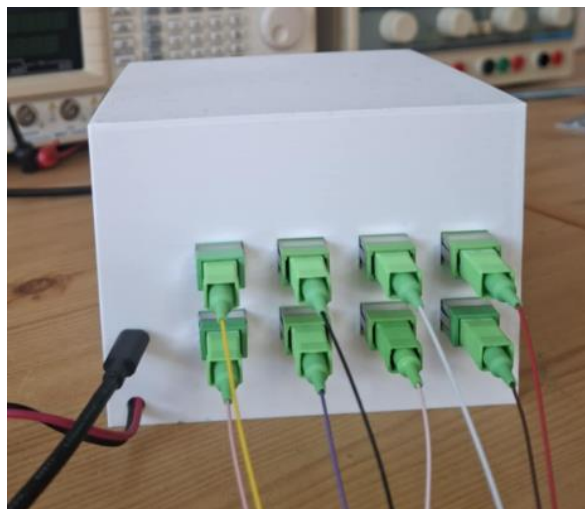


Figure 26 3D printed case..

3.5. Signal Processing and Software Implementation

3.5.1. Peripheral Initialization

Bare-Metal Approach: To ensure full control over the hardware resources and minimize program memory space, the software was developed using a Direct Register Access (Bare-Metal) approach. Standard high-level libraries (such as STM32 HAL) were deliberately omitted. Instead, custom low-level drivers were developed for the GPIO, ADC, and USART peripherals, operating directly on the memory-mapped registers. The system clock was maintained at the default internal 8 MHz frequency to minimize power consumption.

GPIO Configuration: A dedicated GPIO driver was implemented to encapsulate the complex bit-masking operations required for port configuration. Designed initialization function was created, allowing for the configuration of pin modes (Input/Output/Alternate Function), output types, and pull-up/pull-down resistors with a single line of code. This function was used to program all of the peripheral GPIO pins, making the code readable and easy to understand.

ADC Management: The ADC driver manages the precise timing and calibration of the measurement subsystem.

1. **Calibration:** To eliminate offset errors inherent in the manufacturing process, a self-calibration routine is executed at startup. The ADC calculates a 7-bit calibration factor, which is stored in the memory and applied to subsequent measurements to ensure data integrity [69].
2. **Resolution and Timing:** The ADCs are configured for 8-bit resolution. The sampling time was set to 19.5 clock cycles. Adding the fixed conversion time of 12.5 cycles, the total time for a single conversion is 32 clock cycles. With an 8MHz system clock, the conversion time (T_{conv}) of one channel is calculated as:

$$T_{conv} = \frac{32 \text{ cycles}}{8MHz} = 4\mu s \quad (16)$$

3. **Sequence Management:** Two channels are assigned to each ADC peripheral. The driver utilizes the Regular Group Sequencer to convert these channels consecutively and store the results in memory variables.

USART Configuration: The communication interface was configured for a baud rate of 38400bps with an oversampling factor of 16 to ensure noise immunity. The baud rate generation is set by the USART divider (USARTDIV), calculated as [69]:

$$USARTDIV = \frac{f_{CK}}{Baud} = \frac{8000000}{38400} \approx 208.33 \quad (17)$$

The driver implements functions for transmitting single characters and full string messages. A receive function was also implemented for potential future bidirectional control, although it is currently unused.

3.5.2. Code architecture

Code Design: The main application consists of an initialization phase followed by a continuous monitoring loop.

After power-up, the system performs a sequential setup:

1. System Clock configuration.
2. Peripheral initialization (GPIO, USART).
3. ADC Calibration and Startup.

Event Detection Loop: The core logic iterates through all the possible sensor combinations (Truth Table states [Table 17]) to detect footstep events.

- **Polling:** The microcontroller continuously reads the digital status of the comparator inputs.
- **Temporal Filtering (Debouncing):** To prevent false triggers caused by mechanical vibrations or transient noise, a software debouncing algorithm was implemented. After detecting trigger condition, the system waits for 200ms, making some sort of a validation window. After this delay, the check() function re-verifies the sensor state.
- **Action:** If the condition persists (confirming a valid step), the system:
 1. Triggers the ADC conversion to capture the analog intensity of the pressure.
 2. Compiles the logic state and analog data.
 3. Transmits the packet via USART to the user terminal.
- **Visualization:** In addition to raw data, the system transmits an ASCII-based graphical interpretation of the active zone. This feature provides immediate visual feedback on a standard serial terminal, serving as a fallback in case the dedicated PC application is unavailable.

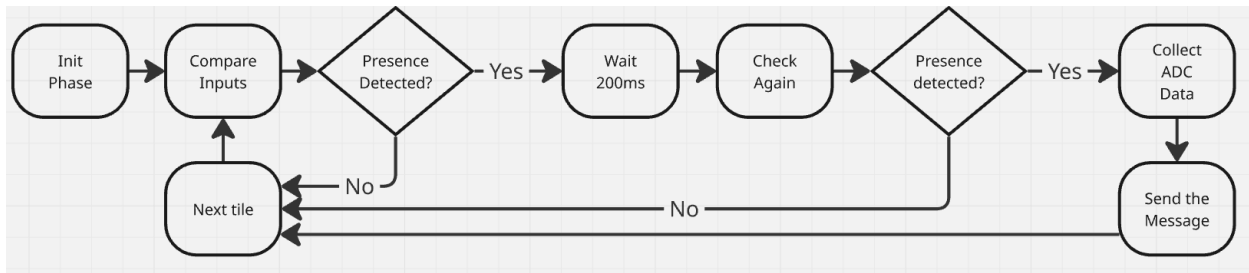


Figure 27 Detection algorithm diagram.

3.5.3. PC User Application

Application Purpose: To visualize the spatial data and monitor the analog parameters in real-time, a dedicated desktop application was developed using the Python programming language. The application serves as a Graphical User Interface (GUI) for the sensor system, allowing for intuitive tracking of the user's position and debugging of the analog signal levels.

Software Architecture: The application relies on the standard Tkinter library for rendering the graphical interface, ensuring cross-platform compatibility and low resource overhead. To handle the asynchronous nature of serial communication without freezing the user interface, a multithreaded architecture was implemented:

1. **Main Thread (GUI Loop):** Handles the rendering of the window, widget updates, and user interactions.
2. **Background Thread (Daemon):** Dedicated to reading the serial port buffer. It uses the serial and threading libraries to continuously listen for incoming data packets from the microcontroller.

Functional Layout: The Graphical User Interface is divided into three functional modules:

- **Spatial Grid Map:** The central element of the dashboard is a dynamic grid representing the physical layout of the room. The grid consists of 15 tiles (arranged in a 3x5 matrix, corresponding to the binary zones). When the microcontroller reports a position (e.g., "A1"), the application analyses this ID and immediately changes the background color of the corresponding tile to red (#ff4c4c), providing a clear visualization of presence.
- **Analog Monitor (ADC):** To observe the raw intensity of the signals, four progress bars display the voltage levels (0–3300 mV) received from the photodiodes.
- **Event History:** A rolling buffer logs the last 5 detected positions, allowing for the analysis of movement sequences.

Data Protocol: The communication operates at a baud rate of 38400 bps. The application implements a parsing algorithm that filters out ANSI escape codes (used for terminal formatting on the microcontroller side) and extracts relevant telemetry data.

- **Position Data:** Lines containing the pos: tag trigger an update of the grid map.
- **Telemetry Data:** Lines containing the ADC tag are parsed to update the voltage bars.
- **Synchronization:** The application detects the "Clear Screen" command sent by the MCU as a reset signal, automatically clearing the map and history to prepare for a new measurement session.

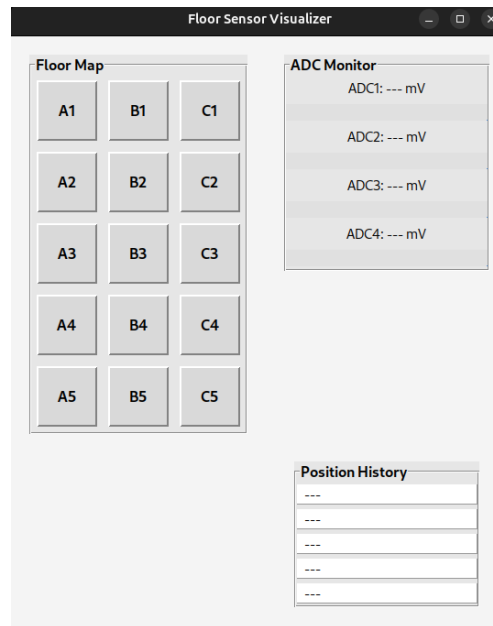


Figure 28 PC application GUI.

4. Results

4.1. Electronic Subsystems Verification

4.1.1. TEC Thermal Stabilization

Before the integrated system tests, a series of measurements were conducted to verify the stability and performance of the designed PCB modules.

The temperature control loop was tested by connecting an ammeter in series with the Peltier element supply. Under nominal conditions (ambient temperature $\approx 22^{\circ}\text{C}$, V_{sense} set point = 2.4V), the control logic exhibited a pulsed operation mode. The transistor opened approximately once per second, allowing a current of 260 mA to flow through the cooler, successfully maintaining the laser temperature stability.

To verify the dynamic response, a disturbance was introduced using a hot air station set to 100°C (applied from a safe distance, by that decreasing the temperature).

- **Response:** The switching frequency increased significantly to counteract the rising temperature.
- **Limit Testing:** When the setpoint was adjusted to tolerate higher temperatures, the duty cycle decreased, but the peak current drawn during the "ON" state exceeded 700mA.

Conclusion: The circuit operates correctly, effectively rejecting thermal disturbances and maintaining the laser diode within safe operating limits.

4.1.2. APC Current Regulation

The Automatic Power Control circuit was characterized by measuring the laser supply current relative to the potentiometer setting (0–10k Ω).

- **Target vs. Measured:** The theoretical design assumed a current of 50mA at 0 Ω potentiometer resistance. The measured value was **34mA**. This difference might be caused by the fact that the monitor photodiode current tests (Table 18) were performed without load, leading to a slight offset in the feedback loop calibration.
- **Control Range:** Increasing the resistance to the maximum value (10k Ω) successfully reduced the current to **0mA**, turning off the laser.
- **Safety Feature:** It was observed that the APC chip enters a latch-off state when the laser is completely disabled; a power cycle is required to reset the driver, which functions as an intended safety mechanism.

4.1.3. Receiver Sensitivity

The calibration of the receiver stage involved setting the reference voltage (V_{ref}) for the comparators. Experimental trials in the complete setup indicated that setting V_{ref} below **700mV** resulted in frequent false alarms due to system noise. Consequently, the threshold was adjusted above this noise floor to ensure reliable binary triggering.

4.2. Mechanical Sensor Optimization

The most critical phase of the research was the optimization of the fiber-optic sensing mechanism. Several mechanical coupling methods were evaluated to maximize sensitivity and repeatability while ensuring durability.

4.2.1. Initial Approaches (Failures and Lessons Learned)

- **Bare Fiber:** Direct interaction with the tight-buffered fiber gave unpredictable results. The ADC readings were irregular, between **700 mV and 1400 mV**, and occasionally showed full scale (1967 mV, limited by the 2V supply minus losses), making it impossible to establish a pattern.
- **Plastic Tube Actuator:** Placing a thin plastic tube over the fiber provided high sensitivity and a strong signal (immediate saturation to max voltage). However, two critical issues were identified:
 1. **Saturation:** The binary response (Signal/No Signal) made it difficult to resolve "ghosting" issues when multiple subjects stood on adjacent fields.
 2. **Destructive Failure:** The concentrated stress caused by the tube damaged the fiber buffer and eventually led to fiber breakage.
- **Foam Interface:** Introducing a foam layer under the tube improved the dynamic range, allowing for a distinction between light and heavy pressure. However, the foam exhibited significant hysteresis and rapid mechanical degradation, leading to changed readings for the same applied weight over time.

4.2.2. Optimized Topology: Self-Microbending

The final and most effective solution involved a "fiber crossing" topology, where the fiber loops were routed to cross over each other. This utilized the self-microbending effect, where one fiber acts as the pressure point for the other. Tests were conducted with a static load of **80 kg**:

- **Single Crossing:** Resulted in a voltage drop range of **0 – 950 mV**. The effect was irregular because of the small surface area of the intersection point relative to the tile size (25×25 cm).
- **Triple Crossing:** Resulted in a voltage drop of **1600 – 1967 mV**. The sensor proved too sensitive, reaching saturation too quickly, which effectively removed the analog information required for advanced algorithms.

- **Double Crossing (Selected):** This configuration offered the optimal balance. Under an 80 kg load, the voltage drop stabilized in the range of **1100 – 1250 mV**. This delivered a reliable, repeatable analog signal with possibility to differentiate between noise, minor objects, and human presence

4.3. Final System Operation

The integration of the Double Crossing fiber topology with the calibrated electronics showed satisfactory results. The final verification confirmed that the system correctly identifies the user's location on the measurement area. Presence is signaled through two channels:

- **Analog/Hardware:** LED diodes connected directly to the comparator outputs provide instantaneous feedback.
- **Digital/Software:** The PC application displays the active zone on a graphical map along with the ADC readings and footstep history.

The test were conducted on the constructed sensing mat shown in Figure 29. The setup consisted of 15 zones, with fibers located as planned in Figure 15. During the tests, user walked on the panels changing the foot location from the center to the boundaries.



Figure 29 Pressure sensing mat setup.

4.3.1. Signal Homogeneity

A key achievement of the mechanical design is the uniformity of the sensor's reading. Localizing the proper tile wasn't the only purpose of the device, measurements showed that when a user applies pressure to the tile containing multiple fibers beneath, the voltage drop read by the ADC ports is nearly the same. The difference between channels was observed to be within the range of $\pm 150\text{mV}$. This confirms that the tight buffer fiber combined with the mesh structure effectively distributes mechanical stress, which is promising indicator for implementing advanced weight-estimation algorithms in the future.

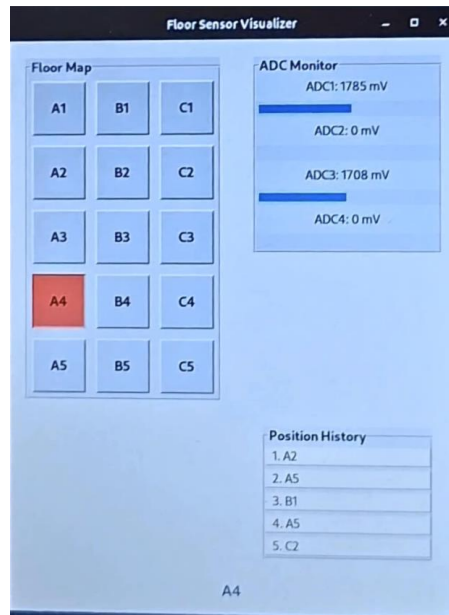


Figure 30 Result of applying pressure to tile A4.

4.3.2. Dynamic Behavior of the System

Response Latency: During the transition between zones (walking across the mat), a brief period of “no result” was observed in the PC application. This behavior is “intentional” and results from the implemented software debouncing algorithm (set to 200ms), which prevents false alarms and variations of output displayed, it leaves some time for a signal to stabilize after changing the tiles. The app delay is irrelevant, as it is less than one second. In the same time the LEDs react instantaneously, this confirms that the app delay is purely a software thing.

Mechanical Relaxation: Rapid movement across the mat showed the limitations of the fibers serving as a sensors. When the user steps off a tile, the optical fiber requires some time to decompress and return to their original shape. Because of that, for a fraction of second, the previous zone may still register as active while the new tile is already pressed or even different zone might get active because of the noise left by previous zone. Although this state is short (similarly as the response latency it is less than 1 second), it affects the footstep tracking history. This suggests applying longer debouncing time, for example 500ms, or applying some material between the fibers and the tile, so the fibers don't get compressed so much and therefore require less time to recover.

Multi-Subject Detection Limitation: The current version of the algorithm uses a simple input comparison, this approach tracks single person effectively, but has difficulties with more subjects. Tests involving two people stepping on different tiles in the same time resulted in displaying wrong location. However, analysis of the ADC reading data proves that the hardware is capable of multi-subject detection. In such scenarios, the voltage discrepancy is clearly visible, the ADC channels corresponding to the fiber which is pressed by two different tiles shows doubled voltage. This shows that the limitation is algorithmic and implementing more complex logic would solve the issue which is a key objective for the future development.

Figure 31 presents the results of applying pressure to tiles A2 (which presses fibers 2 and 3) and B4 (fibers 1, 2 and 3) in the same time. The highlighted area is only A2, meaning the result is wrong (half wrong). However, the ADC readings show that channels 2 and 3 demonstrate twice as high voltage ($\sim 1750\text{mV}$ vs 880mV) as the channel 1, meaning more weight was placed on the fibers 2 and 3. Because for the time of this measurement, the threshold of detection was increased to around 1V, this 880mV on channel 1 “didn’t count” for the processor and the active tile in the application was A2, lowering the threshold below 880mV would result in changing the active tile to B4 with the same ADC readings displayed.

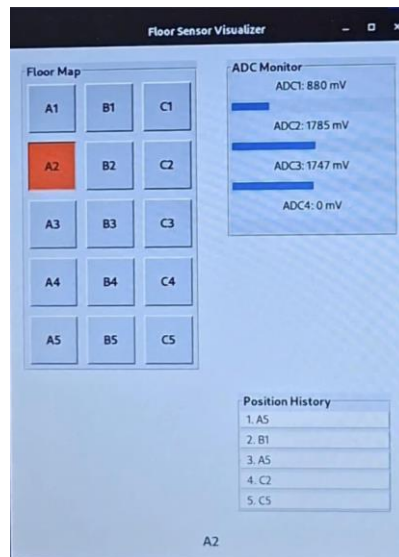


Figure 31 "Ghosting effect" in multi-subject detection.

5. Conclusions

The primary goal of the thesis was to develop a low-cost, fiber-optic based system for human presence detection and localization. Based on the theoretical analysis and experimental validation described in previous chapters, following conclusions can be drawn:

- **Proof of Concept:** Constructed device demonstrates that the optical fiber can be used as an intrinsic sensor and detect pressure, which can be lead to locate person in a properly mapped area.
- **Hardware Efficiency:** The simplified architecture, utilizing a single laser source split into multiple channels, proved to be cost-effective (although not using the cheapest light source). The implementation of the APC and TEC drivers achieved optical stability to perform reliable analog measurements without the need for expensive equipment.
- **Sensing Mechanism:** The experimental phase identifies the most reliable fiber placement method as the “Double Crossing” topology. This method ensured acceptable output range for the same weight measured across all tiles (1100mV - 1250mV).
- **Device Scalability:** The potentiometers present in laser drivers circuits effectively allow for scaling the device for longer/shorter fiber distances, as well as, changing the temperature operating point.
- **Result Visualization:** Python based Graphical User Interface efficiently visualizes the results on user’s PC without the need of being in the same room as the device. The possibility to read the results from the computer terminal or even straight from the PCB diodes increases the safe usage.

Limitations: Despite successful operation, the current prototype exhibits certain limitations:

- **Multi-Subject detection:** Due to the binary logic used in the fiber layout, the presence of multiple people on the mat simultaneously can lead to "ghosting" effects or incorrect localization.
- **Mechanical Durability:** While the rubber mat substrate provides stability, the long-term usage of the optical fibers under repeated mechanical stress requires further lifecycle testing.

Future Development: Future iterations of this project will include:

- Multi-Subject detection algorithm allowing to distinguish more positions in the same time.
- Ethernet communication, allowing the connection of the device to the external optical switch. With use of that, more positions could be checked with the same amount of detectors.
- Introducing more photodiodes, allowing to increase the resolution of the floor map grid.

In conclusion, the developed detection system bridges the gap between expensive industrial sensors and privacy-intrusive cameras, offering a promising foundation for future Ambient Assisted Living (AAL) technologies.

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Appendix

The complete source code, hardware design files, and documentation for the "Smart Floor" system are available in the public GitHub repository.

Repository URL: <https://github.com/MalyRudy2004/Engineering-Thesis>